

CHAPTER

4

Network Topology (Graph Theory)

4.1 INTRODUCTION

The word topology refers to the science of place. In mathematics, topology is a branch of geometry in which figures are considered perfectly elastic.

Network Topology refers to the properties that relate to the geometry of the network (circuit). These properties remain unchanged even if the circuit is bent into any other shape provided that no parts are cut and no new connections are made.

In electrical engineering, solution of network analysis problems involves finding the current through and voltage across different circuit elements. Different laws (like, Ohm's law, Kirchhoff's laws, etc.) have been postulated for simplifying the solution method. However, it is sometimes found that the algebraic equations written by different laws are not independent. On the other hand, the equations formed by network topology method are all independent.

Network topology method has many other merits and can be listed as follows.

1. The graph theory or network topology deals with those properties of networks which do not change with the change in the shape of the networks.
2. All the equations (KCL and KVL) formed by graph theory concept are independent equations.
3. The graph theory concept eases the solution method for solving networks with a large number of nodes and branches.

In this chapter, we will discuss the fundamentals of graph theory (network topology) and their applications for solving network analysis problems.

4.2 GRAPH OF A NETWORK

A linear graph (or simply a graph) is defined as a collection of points called nodes, and line segment called branches, the nodes being joined together by the branches.

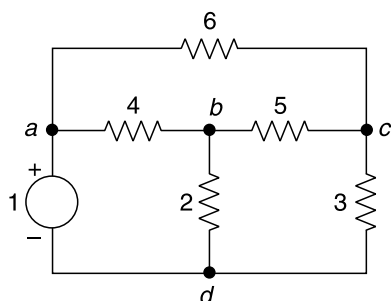


Figure 4.1(a) Circuit

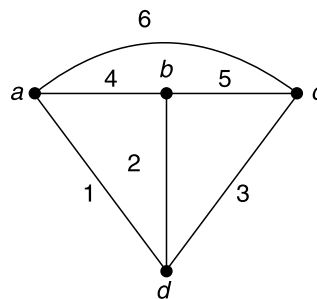


Figure 4.1(b) Graph of the Circuit

While drawing graph of a given network, the following rules are to be noted.

- All passive elements between the nodes are represented by lines.
- The independent current sources and voltage sources are represented by their internal impedances (i.e., current sources by open circuit and voltage sources by short circuit) if they are accompanied by passive element, viz., a shunt admittance in a current source and a series impedance in a voltage source.
- If the sources are not accompanied by passive elements, an arbitrary impedance (say resistance R) or admittance is assumed to accompany the sources and finally, we find the results by letting the impedance $R \rightarrow 0$ or $R \rightarrow \infty$ as the case may be for the current or voltage sources.

4.3 TERMINOLOGY

In order to discuss the more involved methods of circuit analysis, we must define a few basic terms necessary for a clear, concise description of important circuit features.

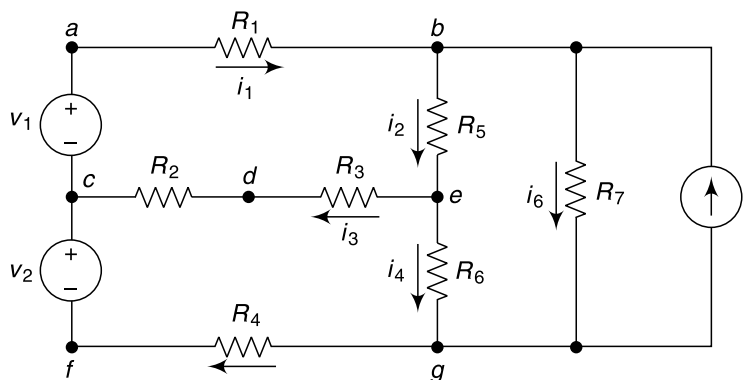


Figure 4.2 Circuit illustrating terminologies

(a) **Node** A node is a point in a circuit where two or more circuit elements join.

Example a, b, c, d, e, f and g

(b) **Essential Node** A node that joins three or more elements.

Example b, c, e and g

(c) **Branch** A branch is a path that connects two nodes.

Example $v_1, R_1, R_2, R_3, v_2, R_4, R_5, R_6, R_7$ and I

(d) **Essential branch** Those paths that connect essential nodes without passing through an essential node.

Example $c-a-b, c-d-e, c-f-g, b-e, e-g, b-g$ (through R_7), and $b-g$ (through I)

(e) **Loop** A loop is a complete path, i.e., it starts at a selected node, traces a set of connected basic circuit elements and returns to the original starting node without passing through any intermediate node more than once.

Example $abedca, abegfca, cdebgfc$, etc.

(f) **Mesh** A mesh is a special type of loop, i.e., it does not contain any other loops within it.

Example $abedca, cdebgfc, gebg$ (through R_7) and $gebg$ (through I)

(g) **Oriented Graph** A graph whose branches are oriented is called a directed or oriented graph.

(h) **Rank of Graph** The rank of a graph is $(n-1)$ where n is the number of nodes or vertices of the graph.

(i) **Planar and Non-planar Graph** A graph is planar if it can be drawn in a plane such that no two branches intersect at a point which is not a node.

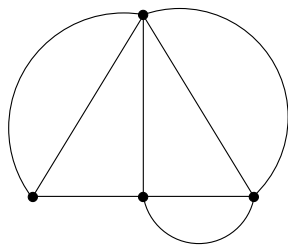


Figure 4.3(a) Planar graph

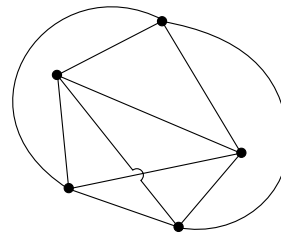


Figure 4.3(b) Non-planar graph

(j) **Subgraph** A subgraph is a subset of the branches and nodes of a graph. The subgraph is said to be proper if it consists of strictly less than all the branches and nodes of the graph.

(k) **Path** A path is a particular sub graph where only two branches are incident at every node except the internal nodes (i.e., starting and finishing nodes). At the internal nodes, only one branch is incident.

In the example in the Fig. 4.3 (c), branches 2, 3, and 4, together with all the four nodes, constitute a path. A graph is connected if there exists a path between any pair of vertices. Otherwise, the graph is disconnected.

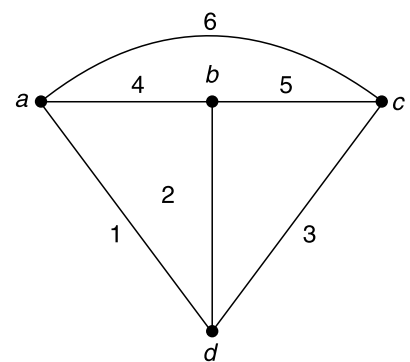


Figure 4.3(c)

4.4 CONCEPT OF TREE

For a given connected graph of a network, a connected subgraph is known as a tree of the graph if the subgraph has all the nodes of the graph without containing any loop.

Twigs The branches of tree are called twigs or tree-branches. The number of branches or twigs, in any selected tree is always one less than the number of nodes, i.e.,

$$\text{Twigs} = (n - 1), \text{ where } n \text{ is the number of nodes of the graph.}$$

For this case, $\text{twigs} = (4 - 1) = 3$ twigs. These are shown by solid lines in Fig. 4.4 (b).

Links and Co-tree If a graph for a network is known and a particular tree is specified, *the remaining branches* are referred to as the **links**. The **collection of links** is called a **co-tree**. So, co-tree is the complement of a tree. These are shown by dotted lines in Fig. 4.4(b).

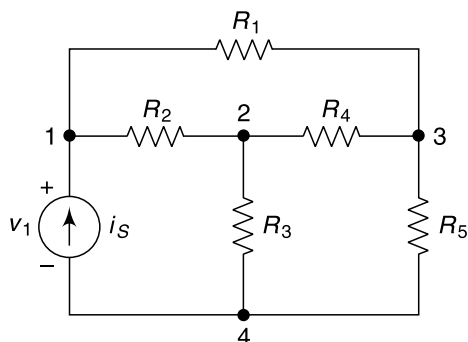


Figure 4.4(a) Circuit

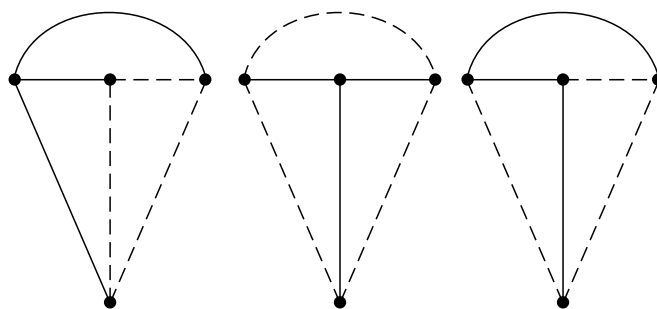


Figure 4.4(b) Trees and links of circuit of Fig. 4.4(a)

The branches of a co-tree may or may not be connected, whereas the branches of a tree are always connected.

To summarize,

Number of nodes in a graph = n

Number of independent voltages = $n - 1$

Number of tree-branches = $n - 1$

Number of links = $L = (\text{Total number of branches}) - (\text{Number of tree-branches})$
 $= b - (n - 1) = b - n + 1$

Total number of branches = $b = L + (n - 1)$

Properties of a Tree

1. In a tree, there exists one and only one path between any pairs of nodes.
2. Every connected graph has at least one tree.
3. A tree contains all the nodes of the graph.
4. There is no closed path in a tree and hence, tree is circuitless.
5. The rank of a tree is $(n - 1)$.

4.5 INCIDENCE MATRIX $[A_a]$

The incidence matrix symbolically describes a network. It also facilitates the testing and identification of the independent variables. Incidence matrix is a matrix which represents a graph **uniquely**.

For a given graph with n nodes and b branches, the complete incidence matrix A_a is a rectangular matrix of order $n \times b$, whose elements have the following values.

$$\text{Number of columns in } [A_a] = \text{Number of branches} = b$$

Number of rows in $[A_a] = \text{Number of nodes} = n$

$A_{ij} = 1$, if branch j is associated with node i and oriented away from node i .

$= -1$, if branch j is associated with node i and oriented towards node i .

$= 0$, if branch j is not associated with node i .

This matrix tells us which branches are incident at which nodes and what are the orientations relative to the nodes.

Example

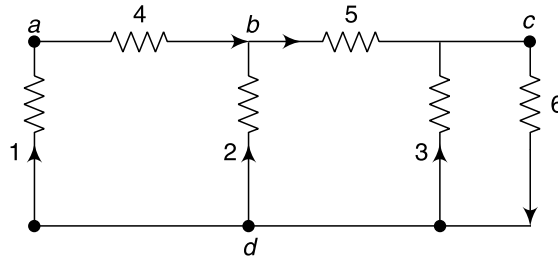


Figure 4.5(a) Network

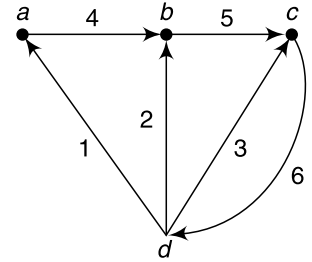


Figure 4.5(b) Graph of network

Incidence matrix A_a

		Branches						
		1	2	3	4	5	6	
Nodes	a	-1	0	0	+1	0	0	} Reduced Incidence Matrix A
	b	0	-1	0	-1	+1	0	
	c	0	0	-1	0	-1	+1	
Reference Node	d	+1	+1	+1	0	0	-1	

4.5.1 Incidence Matrix and KCL

For the graph, shown in Fig. 4.6, Kirchhoff's current law for the branch currents ($i_1, i_2, \dots, i_6$) gives the equations,

$$\begin{aligned} i_1 + i_2 + i_6 &= 0 \\ -i_1 + i_3 - i_5 &= 0 \\ -i_2 - i_3 + i_4 &= 0 \\ -i_4 + i_5 - i_6 &= 0 \end{aligned}$$

In matrix form, these equations can be represented as,

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} i_{b1} \\ i_{b2} \\ i_{b3} \\ i_{b4} \\ i_{b5} \\ i_{b6} \end{bmatrix} = 0$$

or

$$A_a I_b = 0$$

where, A_a is the complete incidence matrix of the graph.

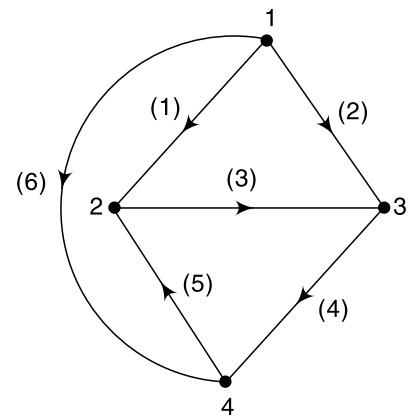


Figure 4.6 Graph illustrating incidence matrix and KLC

Reduced Incidence Matrix $[A]$ The matrix obtained from A_a by eliminating one of the rows is called Reduced Incidence Matrix. In other words, suppression of the datum node (reference node) from the incidence matrix results in reduced incidence matrix.

For the graph shown in Fig. 4.6, reduced incidence matrix is given as,

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 1 & 0 & 0 \end{bmatrix}$$

4.5.2 Incidence Matrix and KVL

For the graph shown in Fig. 4.6, the branch voltages ($v_{b1}, v_{b2}, \dots, v_{b6}$) can be represented in terms of the node voltages ($v_{n1}, v_{n2}, v_{n3}, v_{n4}$) as,

$$\begin{aligned} v_{b1} &= (v_{n1} - v_{n2}), & v_{b2} &= (v_{n1} - v_{n3}), & v_{b3} &= (v_{n2} - v_{n3}), & v_{b4} &= (v_{n3} - v_{n4}), \\ v_{b5} &= (-v_{n1} + v_{n4}), & v_{b6} &= (v_{n1} - v_{n4}), \end{aligned}$$

Thus, the Kirchhoff's voltage law in matrix form can be written as,

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & -1 & 0 & 1 \\ 1 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} V_{n1} \\ V_{n2} \\ V_{n3} \\ V_{n4} \end{bmatrix} = \begin{bmatrix} v_{b1} \\ v_{b2} \\ v_{b3} \\ v_{b4} \\ v_{b5} \\ v_{b6} \end{bmatrix}$$

or

$$A_a^T V_n = V_b$$

4.5.3 Properties of Complete Incidence Matrix

- (i) The sum of the entries in any column is zero.
- (ii) The determinant of the incidence matrix of a closed loop is zero.
- (iii) The rank of incidence matrix of a connected graph is $(n-1)$.

4.6 NUMBER OF POSSIBLE TREES OF A GRAPH

The number of possible trees of a graph, $= \det \{[A] \times [A]^T\}$

where, A is the reduced incidence matrix obtained by eliminating any one row of the complete incidence matrix A_a , and $[A]^T$ is the transpose of the matrix $[A]$.

Example

For the graph shown in Fig. 4.6, the complete incidence matrix is,

$$A_a = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & -1 \end{bmatrix}$$

So, reduced incidence matrix is,

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 1 & 0 & 0 \end{bmatrix}$$

Thus, the number of possible trees of the graph of Fig. 4.6

$$= \det \left\{ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \right\} = \begin{vmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{vmatrix} = 16$$

4.7 TIE-SET MATRIX AND LOOP CURRENTS

Tie-Set A *tie-set* is a set of branches contained in a loop such that each loop contains one link or chord and the remainder are tree branches.

Consider the graph and the tree as shown. This selected tree will result in *three fundamental loops* as we connect each link, in turn to the tree.

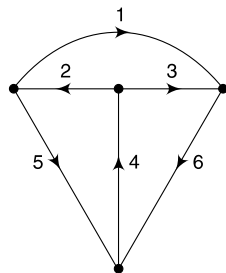


Figure 4.7(a) Graph

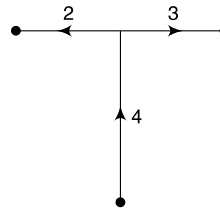


Figure 4.7(b) Tree of the graph

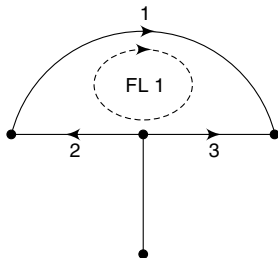


Figure 4.7(c) Loop-1

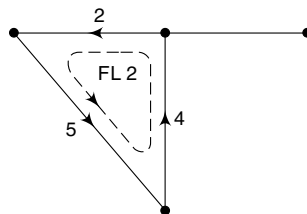


Figure 4.7(d) Loop-2

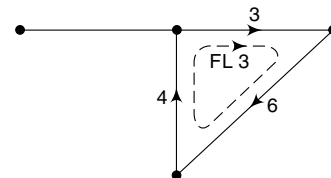


Figure 4.7(e) Loop-3

Fundamental Loop 1 (FL1): Connecting link 1 to the tree.

Fundamental Loop 2 (FL2): Connecting link 5 to the tree.

Fundamental Loop 3 (FL3): Connecting link 6 to the tree.

These sets of branches (1, 2, 3), (2, 4, 5) and (3, 4, 6) form three tie-sets.

4.7.1 Tie-Set Matrix or Loop Incidence Matrix or Circuit Matrix (B_a)

For a given graph having n nodes and b branches, tie-set matrix is a rectangular matrix with b columns and as many rows as there are loops. Its elements have the following values:

$$B_{ij} = \begin{cases} 1, & \text{if branch } j \text{ is in loop } i \text{ and their orientations coincide (i.e., loop current and branch current flow in the same direction);} \\ -1, & \text{if branch } j \text{ is in loop } i \text{ and their orientations do not coincide;} \\ 0, & \text{if branch } j \text{ is not in loop } i. \end{cases}$$

Example

For the graph shown in Fig. 4.8(a) and tree selected in Fig. 4.8(b), the tie-set matrix is written as follows. The entries in the Tie-set schedule are given as +1 or -1 if the branch current is in the same direction as the link current or not. If the branch current does not depend on the link current, then entry is zero.

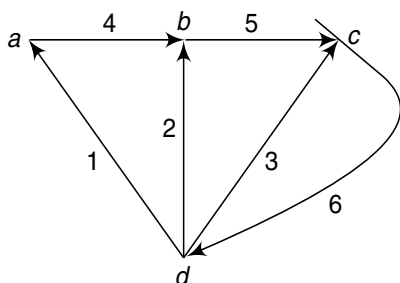


Figure 4.8(a) Graph

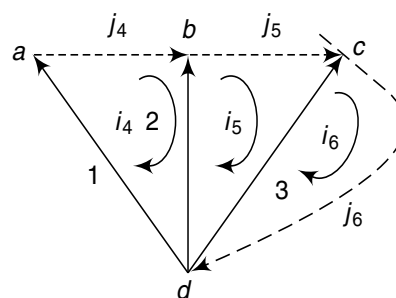


Figure 4.8(b) Formation of loops

Links (j)	Branch no. (i)					
	1	2	3	4	5	6
4	1	-1	0	1	0	0
5	0	1	-1	0	1	0
6	0	0	1	0	0	1

4.7.2 Tie-Set Matrix and KVL

For the graph shown in Fig. 4.7(a) and three loops shown in Fig. 4.7(c), (d) and (e), three fundamental mesh KVL equations can be written as follows.

For Fundamental Loop 1 (FL1): $v_{b1} - v_{b3} + v_{b2} = 0$

For Fundamental Loop 2 (FL2): $v_{b2} + v_{b4} + v_{b5} = 0$

For Fundamental Loop 3 (FL3): $v_{b3} + v_{b6} + v_{b4} = 0$

These equations in matrix form is written as,

$$\begin{bmatrix} 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_{b1} \\ v_{b2} \\ v_{b3} \\ v_{b4} \\ v_{b5} \\ v_{b6} \end{bmatrix} = 0$$

or

$$B_a V_b = 0$$

4.7.3 Tie-Set Matrix and KCL

For the graph shown in Fig. 4.7(a) and three loops shown in Fig. 4.7(c), (d) and (e), the branch currents ($i_{b1}, i_{b2}, \dots, i_{b6}$) can be represented in terms of the loop currents (I_{L1}, I_{L2}, I_{L3}) as,

$$i_{b1} = I_{L1}, \quad i_{b2} = (I_{L1} + I_{L2}), \quad i_{b3} = (-I_{L1} + I_{L3}), \quad i_{b4} = (I_{L2} + I_{L3}), \quad i_{b5} = I_{L2}, \quad i_{b6} = I_{L3}$$

In matrix form, these equations can be written as,

$$\begin{bmatrix} i_{b1} \\ i_{b2} \\ i_{b3} \\ i_{b4} \\ i_{b5} \\ i_{b6} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_{L1} \\ I_{L2} \\ I_{L3} \end{bmatrix}$$

or

$$I_b = B_a^T I_L$$

4.8 CUT-SET MATRIX AND NODE-PAIR POTENTIAL

Cut-Set A cut-set is a minimum set of elements that when cut, or removed, separates the graph into two groups of nodes. A cut-set is a **minimum set of branches** of a connected graph, such that the removal of these branches from the graph **reduces the rank of the graph by one**.

In other words, for a given connected graph (G), a set of branches (C) is defined as a cut-set if and only if:

- (i) the removal of all the branches of C results in an unconnected graph.
- (ii) the removal of all but one of the branches of C leaves the graph still connected.

Example

Consider the graph shown in Fig. 4.9(a). The rank of the graph is 3. The removal of branches 1 and 3 reduces the graph into two connected subgraphs as shown in Fig. 4.9(b).

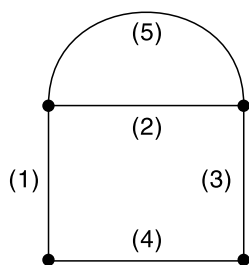


Figure 4.9(a) Graph

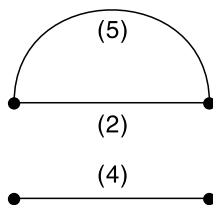


Figure 4.9(b) Subgraphs with removal of 1 and 3

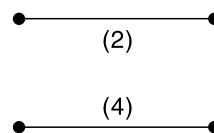


Figure 4.9(c) Subgraphs with removal of 1, 3 and 5

The rank of the graph of Fig. 4.9(a) = $(4 - 1) = 3$

The rank of the graph of Fig. 4.9(b) = Addition of the ranks of the subgraphs = $(1 + 1) = 2$

So, branches [1, 3] may be a cut-set.

Also, removal of the branches 1, 3 and 5 reduces the graph into two connected subgraphs as shown in Fig. 4.9(c) and the rank becomes 2. So, [1, 3, 5] may also be a cut-set.

As cut-set is the minimum set of branches and [1, 3] is a subset of [1, 3, 5], so [1, 3] is the cut-set, [1, 3, 5] is not a cut-set.

4.8.1 Fundamental Cut-Set

A fundamental cut-set (FCS) is a cut-set that cuts or contains **one and only one tree branch**. Therefore, for a given tree, the number of fundamental cut-sets will be equal to the number of twigs.

4.8.2 Procedure for Finding the Fundamental Cut-sets

1. First, select a tree of the given graph.
2. Focus on a tree branch (b_k).
3. Check whether removing this tree branch (b_k) from the tree disconnects the tree into two separate parts.
4. All the links which go from one part of this disconnected tree to the other, together with the tree branch (b_k) forms a fundamental cut-set.

Following this procedure, the fundamental cut-sets for the graph of Fig. 4.10 will be

f -cut-set – 1: [1, 2, 6];

f -cut-set – 2: [2, 3, 5, 6];

f -cut-set – 3: [4, 5, 6]

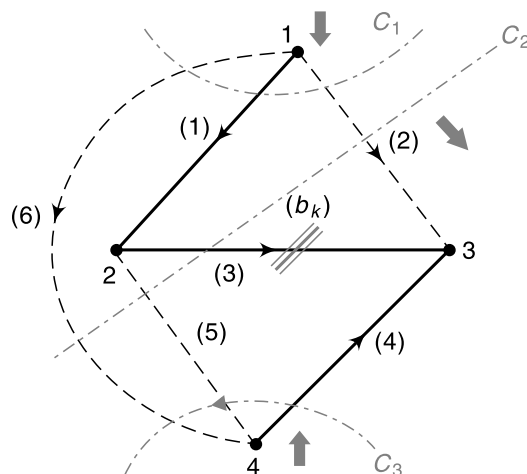


Figure 4.10 Graph illustrating fundamental cut-set

4.8.3 Properties of Cut-Set

1. A cut-set divides the set of nodes into two subsets.
2. Each fundamental cut-set contains one tree-branch, the remaining elements being links.
3. Each branch of the cut-set has one of its terminals incident at a node in one subset and its other terminal at a node in the other subset.

4. A cut-set is oriented by selecting an orientation from one of the two parts to the other. Generally, the direction of cut-set is chosen same as the direction of the tree branch.

4.8.4 Cut-Set Matrix (Q_C)

For a given graph, a cut-set matrix (Q_C) is defined as a rectangular matrix whose rows correspond to cut-sets and columns correspond to the branches of the graph. Its elements have the following values:

$$\begin{aligned} Q_{ij} &= 1, \text{ if branch } j \text{ is in the cut-set } i \text{ and the orientations coincide.} \\ &= -1, \text{ if branch } j \text{ is in the cut-set } i \text{ and the orientations do not coincide.} \\ &= 0, \text{ if branch } j \text{ is not in the cut-set } i. \end{aligned}$$

Example

For the graph shown in Fig. 4.10, fundamental cut-sets have been identified as follows.

$$\begin{aligned} f\text{-cut-set} - 1: & [1, 2, 6]; \\ f\text{-cut-set} - 2: & [2, 3, 5, 6]; \\ f\text{-cut-set} - 3: & [4, 5, 6] \end{aligned}$$

So, the cut-set matrix is written as,

f-cut-sets	Branch no:					
	1	2	3	4	5	6
1	1	1	0	0	0	1
2	0	1	1	0	1	1
3	0	0	0	1	-1	-1

4.8.5 Cut-Set Matrix and KVL

By cut-set schedule, the branch voltages can be expressed in terms of the tree-branch voltages.

A cut-set consists of *one and only one* branch of the tree together with any links which must be cut to divide the network into two parts. A set of fundamental cut-sets includes those cut-sets which are obtained by applying cut-set division for each of the branches of the network tree.

Consider the following graph.

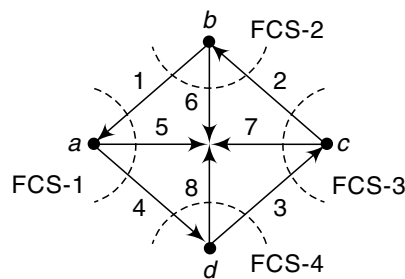


Figure 4.11(a) Graph

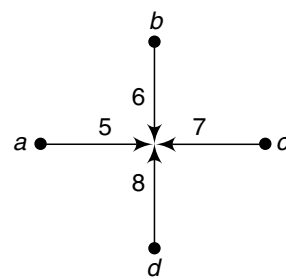


Figure 4.11(b) Tree

Applying cut-sets at nodes a , b , c , d , which are the fundamental cut-sets (FCS), we can write the cut-set schedule as follows.

		1	2	3	4	5	6	7	8
FCS-1 →	a	-1	0	0	1	1	0	0	0
FCS-2 →	b	1	-1	0	0	0	1	0	0
FCS-3 →	c	0	1	1	0	0	0	1	0
FCS-4 →	d	0	0	-1	-1	0	0	0	1

The tree-branch voltages are $[v_{t5}, v_{t6}, v_{t7}, v_{t8}]$ and the branch voltages are $[V_{b1}, V_{b2}, \dots, V_{b8}]$ and the relationship between tree-branch voltages and branch voltages are:

$$\begin{aligned} V_{b1} &= -v_{t5} + v_{t6} & V_{b5} &= v_{t5} \\ V_{b2} &= -v_{t6} + v_{t7} & V_{b6} &= v_{t6} \\ V_{b3} &= v_{t7} - v_{t8} & V_{b7} &= v_{t7} \\ V_{b4} &= v_{t5} - v_{t8} & V_{b8} &= v_{t8} \end{aligned}$$

The above equations can be related by using the cut-set schedule as:

$$\begin{bmatrix} V_{b1} \\ V_{b2} \\ V_{b3} \\ V_{b4} \\ V_{b5} \\ V_{b6} \\ V_{b7} \\ V_{b8} \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_{t5} \\ v_{t6} \\ v_{t7} \\ v_{t8} \end{bmatrix}$$

or

$$\boxed{V_b = Q_c^T V_t}$$

4.8.6 Cut-Set Matrix and KCL

For the graph of Fig. 4.11, writing Kirchhoff's current laws for the nodes, the branch currents can be expressed as,

$$\begin{aligned} \text{Node } a: \quad & -i_{b1} + i_{b4} + i_{b5} = 0 \\ \text{Node } b: \quad & i_{b1} - i_{b2} + i_{b6} = 0 \\ \text{Node } c: \quad & i_{b2} + i_{b3} + i_{b7} = 0 \\ \text{Node } d: \quad & -i_{b3} - i_{b4} + i_{b8} = 0 \end{aligned}$$

In matrix form, they can be written as,

$$\begin{bmatrix} -1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{b1} \\ i_{b2} \\ i_{b3} \\ i_{b4} \\ i_{b5} \\ i_{b6} \\ i_{b7} \\ i_{b8} \end{bmatrix} = 0$$

or

$$\boxed{Q_c I_b = 0}$$

There is a cut-set matrix for a given tree. If a graph contains more than one tree, there will be as many numbers of cut-set matrices as the number of tree of the graph.

To summarize, KVL and KCL equations in three matrix forms are given below.

Matrix	KCL	KVL
Incidence Matrix (A_a)	$A_a \times I_b = 0$	$V_b = A_a^T \times V_n$
Tie-Set Matrix (B_a)	$I_b = B_a^T \times I_L$	$B_a \times V_b = 0$
Cut-Set Matrix (Q_C)	$Q_C \times I_b = 0$	$V_b = Q_C^T \times V_t$

4.9 FORMULATION OF NETWORK EQUILIBRIUM EQUATIONS

The network equilibrium equations are a set of equations that completely and uniquely determine the state of a network at any instant of time. These equations are written in terms of suitably chosen current variables or voltage variables.

These equations will be unique if the number of independent variables be equal to the number of independent equations.

Number of Independent Variables or Equations = $b - (n - 1)$; for loop method of analysis
= $(n - 1)$; for node method of analysis.

The equations for a network can be formed in either of the two methods as given below.

1. Through a set of voltage law equations in which the currents are the independent variables (Loop-Basis Method);
2. Through a set of current law equations in which the node-pair voltages are the independent variables (Node-Basis Method).

4.9.1 Formulation of Network Equations on Loop Basis

Steps

1. Draw the **directed graph** of the network selecting the direction of assumed current flow to coincide for current sources.
2. Select a tree of the graph.
3. Place all **voltage sources in the tree** and all **current sources in the co-tree**.
4. Place all control-voltage branches for voltage-controlled dependent sources in the tree and all control-current branches for current-controlled dependent sources in the co-tree, if possible.
5. Add one link to the tree, creating a fundamental loop, and write a KVL equation for this fundamental loop (FL). Repeat for each additional link until $L (= b - n + 1)$ mesh equations are obtained in the form $B_a \times V_b = 0$.
6. The current sources in the cotree, if present, will provide the constraint equations.
7. The KCL equations are obtained by representing the branch currents in terms of loop currents in the form $I_b = B_a^T \times I_L$.
8. For each branch, the relationship between the voltage and current is obtained from Ohm's law ($V = RI$).
9. Finally, the equilibrium equations are obtained in terms of loop currents by suitable substitution of the equations obtained in Steps 5 to 8.

4.9.2 Formulation of Network Equations on Node Basis

Steps

1. Draw a directed graph of the circuit under considerations, selecting the directions of assumed current flow to coincide for current sources.
2. Select the tree of the graph so that current *sources are in the co-tree and the voltage sources are within the tree, if possible*. Also, if possible, select the tree so that *at least two branches of the tree are incident at the reference node*.
3. Identify $(n - 1)$ fundamental cut-sets (FCS) and draw the FCS lines.
4. Write the $(n - 1)$ FCS KCL equations in the form $A_a \times I_b = 0$ or $Q_C \times I_b = 0$.
5. Obtain each of the branch currents in terms of node voltages in the form $V_b = A_a^T \times V_n$ or, $V_b = Q_C^T \times V_r$.
6. For each branch, the relationship between the voltage and current is obtained from Ohm's law ($V = RI$).
7. Substitute the equations of step 6 into the KVL equations of step 5 and finally into the KCL equations of step 4, thus obtaining the $(n - 1)$ independent node voltage equations.

4.9.3 Generalized Equations in Matrix Forms for Circuits Having Sources

A general branch consisting of a voltage source V_s and a current source I_s is shown in Fig. 4.12.

Here, the branch current is $(I_b + I_s)$ and the branch voltage is $(V_b + V_s)$.

Without sources, the KCL and KVL equations are:

$$A_a \times I_b = 0 \quad (4.1)$$

$$I_b = B_a^T \times I_L \quad (4.2)$$

$$Q_C \times I_b = 0 \quad (4.3)$$

and

$$V_b = A_a^T \times V_n \quad (4.4)$$

$$B_a \times V_b = 0 \quad (4.5)$$

$$V_b = Q_C^T \times V_t \quad (4.6)$$

With the sources, the KCL and KVL equations are modified as,

$$A_a I_b + A_a I_s = 0 \quad (4.7)$$

$$I_b + I_s = B_a^T I_L \quad (4.8)$$

$$Q_C I_b + Q_C I_s = 0 \quad (4.9)$$

and

$$V_b + V_s = A_a^T V_n \quad (4.10)$$

$$B_a V_b + B_a V_s = 0 \quad (4.11)$$

$$V_b + V_s = Q_C^T V_t \quad (4.12)$$

The branch voltage-current relations for the passive network elements are written in matrix form as,

$$V_b = Z_b I_b \quad (4.13)$$

and

$$I_b = Y_b V_b \quad (4.14)$$

where, Z_b is the *branch impedance matrix* and Y_b is the *branch admittance matrix*, both of the order $b \times b$. On the basis of these equations the general equations can be written in terms of three matrices as follows.

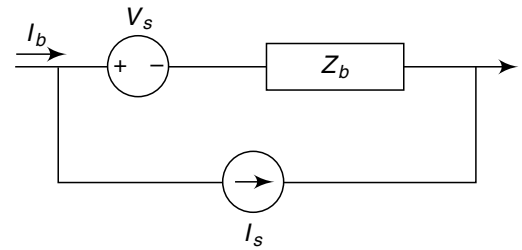


Figure 4.12

Node Equations From equation (4.7),

$$A_a I_s = -A_a I_b = -A_a Y_b V_b = -A_a Y_b (A_a^T V_n - V_s) \quad \{\text{by equation (4.10)}\}$$

or

$$A_a Y_b A_a^T V_n = A_a Y_b V_s - A_a I_s = A_a [Y_b V_s - I_s]$$

or

$$\boxed{Y V_n = A [Y_b V_s - I_s]}$$

where, Y is called the *nodal admittance matrix* of the order of $(n-1) \times (n-1)$. The above equation represents a set of $(n-1)$ number of equations, known as *node equations*.

In case of node analysis, one node is taken as the datum node and potential of that node is zero. Consequently, the complete incident matrix becomes reduced incidence matrix. Thus, the node equations become

$$\boxed{Y V_n = A [Y_b V_s - I_s]}$$

where, $Y = A Y_b A^T$ is called the *nodal admittance matrix* of the order of $(n-1) \times (n-1)$. The above equation represents a set of $(n-1)$ number of equations, known as *node equations*.

Mesh Equations From equation (4.11),

$$B_a V_s = -B_a V_b = -B_a Z_b I_b = -B_a Z_b (B_a^T I_L - I_s) \quad \{\text{by equation (4.8)}\}$$

or

$$B_a Z_b B_a^T I_L = B_a [Z_b I_s - V_s]$$

or

$$\boxed{Z L_L = B_a [Z_b I_s - V_s]}$$

where, Z is the *loop-impedance matrix* of the order of $(b-n+1) \times (b-n+1)$. The above equation represents a set of $(b-n+1)$ number of equations, known as *mesh* or *loop equations*.

Cut-set Equations From equation (4.8),

$$Q_c I_s = -Q_c I_b = -Q_c Y_b V_b = -Q_c Y_b (Q_c^T V_t - V_s) \quad \{\text{by equation (4.12)}\}$$

or

$$Q_c Y_b Q_c^T V_t = Q_c [Y_b V_s - I_s]$$

or

$$\boxed{Y_c V_t = Q_c [Y_b V_s - I_s]}$$

where, Y_c is the *cut-set admittance matrix* of the order of $(n-1) \times (n-1)$ and the set of $(n-1)$ equations represented by the above equation is known as *cut-set equations*.

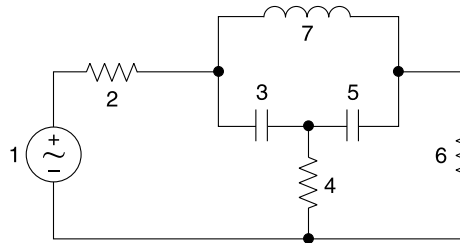
4.10 SOLUTION OF EQUILIBRIUM EQUATIONS

There are two methods of solving equilibrium equations given as follows.

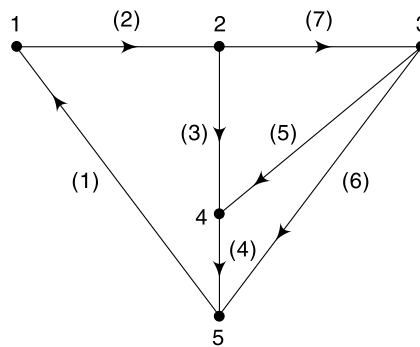
- (i) *Elimination method*: by eliminating variables until an equation with a single variable is achieved, and then by the method of substitution.
- (ii) *Determinant method*: by the method known as Cramer's rule.

SOLVED PROBLEMS

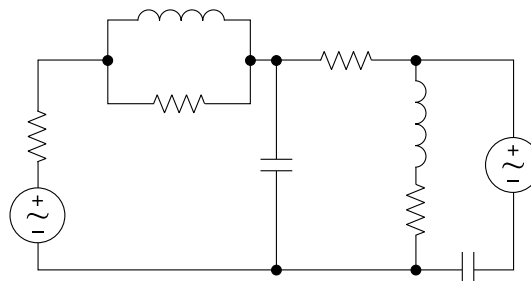
4.1 Draw the graph of the network shown in the figure.



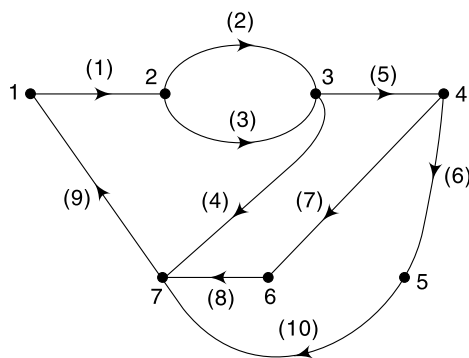
Solution The graph of the network is shown below.



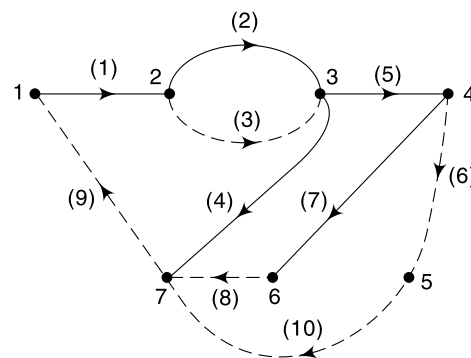
4.2 From the figure, make the graph and find one tree. How many mesh currents are required for solving the network? Find the number of possible trees.



Solution The graph of the network is shown below. One tree of the graph is shown.



Graph of the network



Tree of the graph

The complete incidence matrix is obtained as,

	<i>Nodes</i>	<i>Branches</i>									
		<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>
$A_a =$	1	1	0	0	0	0	0	0	0	-1	0
	2	-1	1	1	0	0	0	0	0	0	0
	3	0	-1	-1	1	1	0	0	0	0	0
	4	0	0	0	0	-1	1	1	0	0	0
	5	0	0	0	0	0	-1	0	0	0	1
	6	0	0	0	0	0	0	-1	1	0	0
	7	0	0	0	-1	0	0	0	-1	1	-1

Reduced incidence matrix becomes,

	<i>Nodes</i>	<i>Branches</i>									
		<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>
$A =$	1	1	0	0	0	0	0	0	0	-1	0
	2	-1	1	1	0	0	0	0	0	0	0
	3	0	-1	-1	1	1	0	0	0	0	0
	4	0	0	0	0	-1	1	1	0	0	0
	5	0	0	0	0	0	-1	0	0	0	1
	6	0	0	0	0	0	0	-1	1	0	0

Hence the number of possible trees is,

$$\begin{aligned}
 n &= \det \left[\begin{array}{c} \left[\begin{array}{cccccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \end{array} \right] \left[\begin{array}{cccccc} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array} \right] \end{array} \right] \\
 &= \det \left[\begin{array}{cccccc} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 3 & -2 & 0 & 0 & 0 \\ 0 & -2 & 4 & 0 & 0 & 0 \\ 0 & 0 & -1 & 3 & -1 & -1 \\ 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & -1 & 0 & 2 \end{array} \right] \\
 \Rightarrow n &= 12 \quad \text{Ans.}
 \end{aligned}$$

4.3 Branch current and loop current relations are expressed in matrix form as,

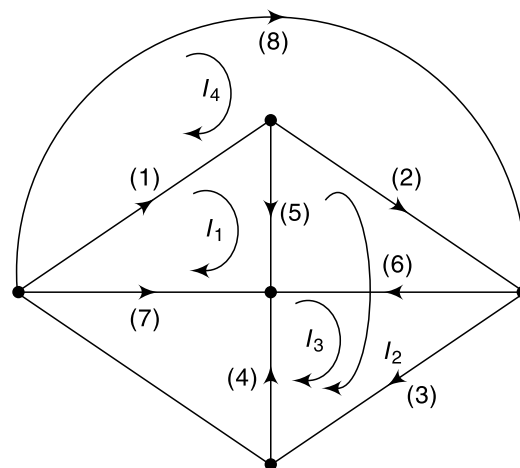
$$\begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \\ i_7 \\ i_8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix}$$

Draw the oriented graph.

Solution We know that, $[I_b] = [B_a]^T [I_L]$. So, the tie-set matrix, here, is,

Loop or Link Currents	Branches							
	1	2	3	4	5	6	7	8
$B_a =$	1	0	0	0	1	0	-1	0
2	0	1	1	1	-1	0	0	0
3	0	0	1	1	0	-1	0	0
4	-1	-1	0	0	0	0	0	1

So, the graph consists of four loops and eight branches. Loop 1 consists of branch 1, 5 and 7. The orientations are given following the sign +1 or -1. Following the procedure, the complete oriented graph is shown below.



4.4 The fundamental cut-set matrix is given as,

Twigs				Links		
1	2	3	4	5	6	7
1	0	0	0	-1	0	0
0	1	0	0	1	0	1
0	0	1	0	0	1	1
0	0	0	1	0	1	0

Draw the oriented graph of the network.

Solution The graph has seven branches and four fundamental cut-sets:

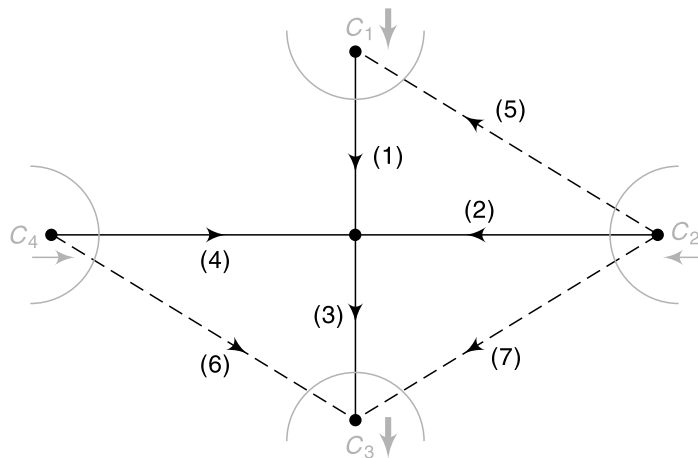
Cut-set-1: [1, 5]

Cut-set-2: [2, 5, 7]

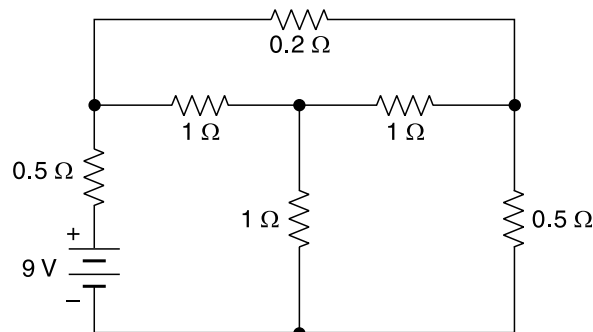
Cut-set-3: [3, 6, 7]

Cut-set-4: [4, 6]

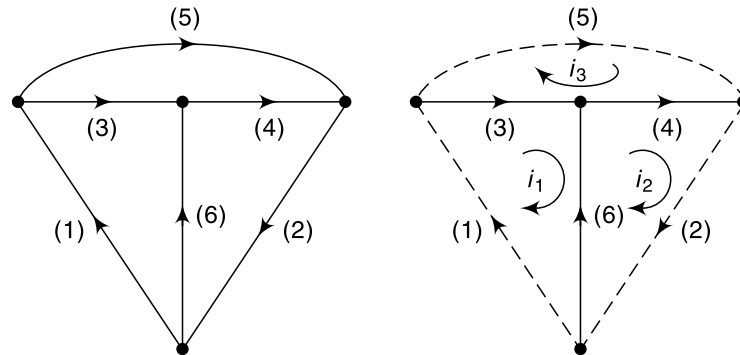
So, the oriented graph is as shown in figure.



- 4.5 (a) For the network of the figure, draw the graph and write a tie-set schedule. Using the tie-set schedule obtain the loop equations and find the currents in all branches.
- (b) For the network of (a), write a cut-set schedule, obtain nodal equations and find branch currents.



Solution The graph and one tree are shown in figure.



The tie-set matrix,

$$B_a = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & -1 & 1 & 0 \end{bmatrix}$$

Branch impedance matrix is,

$$Z_b = \begin{bmatrix} 0.5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Thus,

$$[B_a][Z_b] = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0.5 & 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & -1 & 0.2 & 0 \end{bmatrix}$$

$$\therefore [B_a][Z_b][B_a]^T = \begin{bmatrix} 0.5 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0.5 & 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & -1 & 0.2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2.5 & -1 & -1 \\ -1 & 2.5 & -1 \\ -1 & -1 & 2.2 \end{bmatrix}$$

Now,

$$-[B_a][V_s] = -\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} -9 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \\ 0 \end{bmatrix}$$

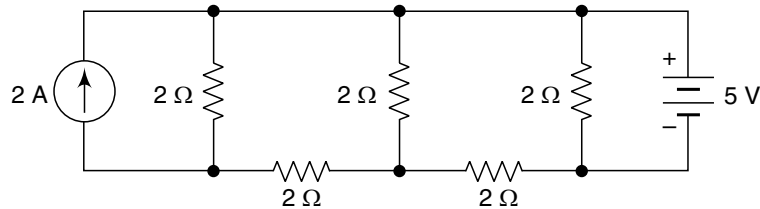
So, the loop equations are,

$$\begin{bmatrix} 2.5 & -1 & -1 \\ -1 & 2.5 & -1 \\ -1 & -1 & 2.2 \end{bmatrix} \times \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \\ 0 \end{bmatrix}$$

Solving three equations,

$$i_1 = 8.9 \text{ A}, i_2 = 6.33 \text{ A}, i_3 = 6.92 \text{ A} \quad \text{Ans.}$$

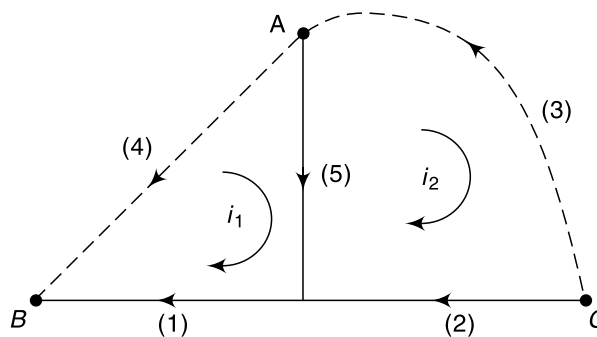
- 4.6 The figure shows a d.c. network. (a) Draw a graph of the network. Which elements are not included in the graph and why? (b) Write a loop incidence matrix and use it to obtain loop equations. (c) Find branch currents.



Solution

- (a) The graph is shown below.

The 2Ω resistor in parallel with voltage source and the 2 A current source have not been included in the graph. This is because of the reason that passive elements in parallel with a voltage source are not included in graph and the current source in parallel with a passive element is open-circuited while drawing graph.



(b) The tie-set matrix for the tree chosen is,

$$B_a = \begin{bmatrix} 1 & 0 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 & -1 \end{bmatrix}$$

Branch impedance matrix is,

$$Z_b = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\begin{aligned} B_a Z_b B_a^T &= \begin{bmatrix} 1 & 0 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & -1 \\ -1 & 0 \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 & 0 & -2 & 2 \\ 0 & 2 & 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & -1 \\ -1 & 0 \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 6 & -2 \\ -2 & 4 \end{bmatrix} \end{aligned}$$

Now,

$$B_a Z_b I_s - B_a V_s = \begin{bmatrix} 2 & 0 & 0 & -2 & 2 \\ 0 & 2 & 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -5 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ -5 \end{bmatrix}$$

So, the loop equations are,

$$\begin{bmatrix} 6 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 4 \\ -5 \end{bmatrix}$$

Solving these equations, $i_1 = 0.3 \text{ A}$, $i_2 = -1.1 \text{ A}$ *Ans.*

(c) Putting these values, the branch voltages are

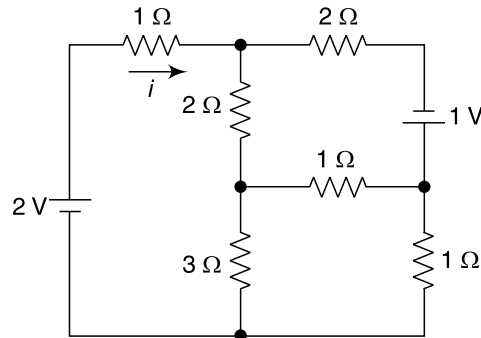
$$V_1 = 2 \times i_1 = 0.6 \text{ V}, V_2 = 2 \times i_2 = -2.2 \text{ V}, V_3 = -5 \text{ V}, V_4 = -2 \times i_1 + 4 = 3.4 \text{ V}, V_5 = 2.8 \text{ V} \quad \text{Ans.}$$

Thus, the branch currents are

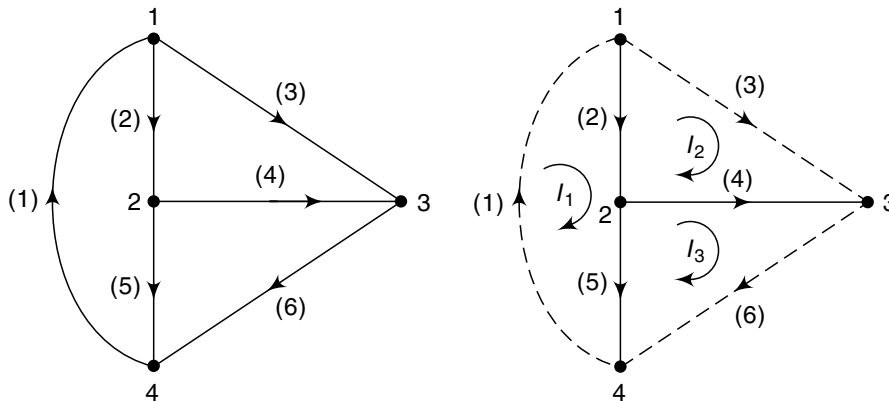
$$I_{AB} = \frac{3.4}{2} = 1.7 \text{ A}, I_{AD} = \frac{2.8}{2} = 1.4 \text{ A}, I_{AC} = \frac{5}{2} = 2.5 \text{ A}, I_{DB} = \frac{0.6}{2} = 0.3 \text{ A}, I_{DC} = \frac{2.2}{2} = 1.1 \text{ A}$$

So, the current supplied by the battery = $(1.7 + 1.4 + 2.5 - 2) = 4.6 \text{ A}$ *Ans.*

- 4.7 For the network shown in the figure, draw the oriented graph and obtain the tie-set matrix. Use this matrix to calculate i .



Solution The oriented graph and any one tree are shown.



The tie-set matrix is given as,

$$B_a = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 \end{bmatrix}$$

The branch impedance matrix,

$$Z_b = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\therefore B_a Z_b = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 0 & 3 & 0 \\ 0 & -2 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -3 & 1 \end{bmatrix}$$

$$\therefore B_a Z_b B_a^T = \begin{bmatrix} 1 & 2 & 0 & 0 & 3 & 0 \\ 0 & -2 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & -2 & -3 \\ -2 & 5 & -1 \\ -3 & -1 & 5 \end{bmatrix}$$

$$\text{Now, } -B_a V_s = - \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = - \begin{bmatrix} -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

So, the loop equations become,

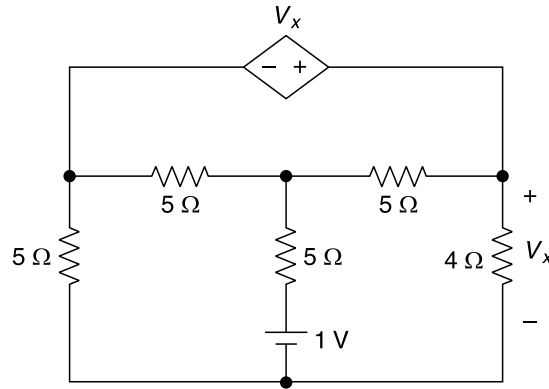
$$\begin{bmatrix} 6 & -2 & -3 \\ -2 & 5 & -1 \\ -3 & -1 & 5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Solving for I_1 ,

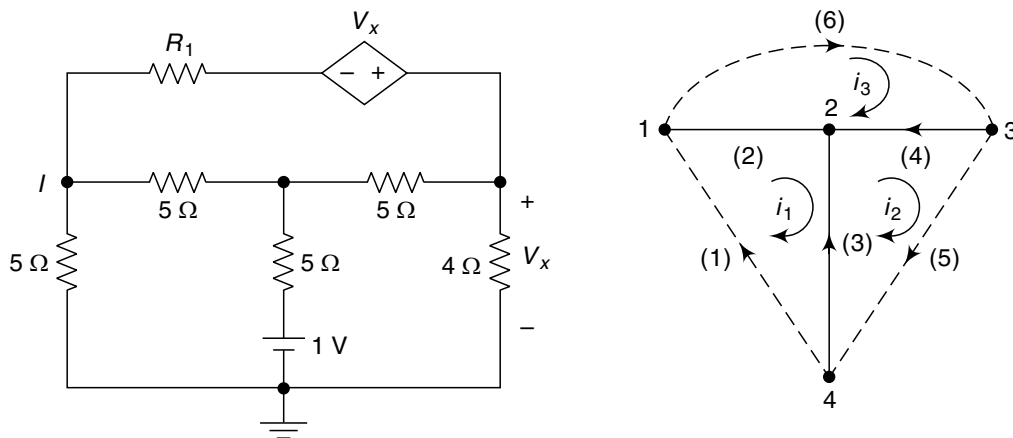
$$I_1 = \frac{\begin{vmatrix} 2 & -2 & -3 \\ 1 & 5 & -1 \\ 0 & -1 & 5 \end{vmatrix}}{\begin{vmatrix} 6 & -2 & -3 \\ -2 & 5 & -1 \\ -3 & -1 & 5 \end{vmatrix}} = 0.91 \text{ A}$$

$$\therefore i_1 = 0.91 \text{ A}$$

- 4.8 The circuit of the figure contains a voltage controlled voltage source. For this circuit, draw the oriented graph. By selecting a proper tree obtain the tie-set matrix and hence calculate the voltage, V_x .



Solution Since the controlled voltage source is not accompanied by any passive element, we will consider a resistance R_1 in series with the controlled voltage source, and finally let $R_1 \rightarrow 0$.



The graph of the network is shown with one tree.
The tie-set matrix is,

$$B_a = \begin{bmatrix} 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

The branch impedance matrix,

$$Z_b = \begin{bmatrix} 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & R_1 \end{bmatrix}$$

$$\therefore B_a Z_b = \begin{bmatrix} 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & R_1 \end{bmatrix} = \begin{bmatrix} 5 & 5 & -5 & 0 & 0 & 0 \\ 0 & 0 & 5 & -5 & 4 & 0 \\ 0 & -5 & 0 & 5 & 0 & R_1 \end{bmatrix}$$

$$\therefore B_a Z_b B_a^T = \begin{bmatrix} 5 & 5 & -5 & 0 & 0 & 0 \\ 0 & 0 & 5 & -5 & 4 & 0 \\ 0 & -5 & 0 & 5 & 0 & R_1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 15 & -5 & -5 \\ -5 & 14 & -5 \\ -5 & -5 & (10+R_1) \end{bmatrix}$$

$$\text{Now, } -B_a V_s = - \begin{bmatrix} 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ 0 \\ -V_x \end{bmatrix} = - \begin{bmatrix} 1 \\ -1 \\ -V_x \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ V_x \end{bmatrix}$$

So, the loop equations become,

$$\begin{bmatrix} 15 & -5 & -5 \\ -5 & 14 & -5 \\ -5 & -5 & (10+R_1) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ V_x \end{bmatrix}$$

With $R_1 \rightarrow 0$ and $V_x = 4I_2$, the equations reduce to,

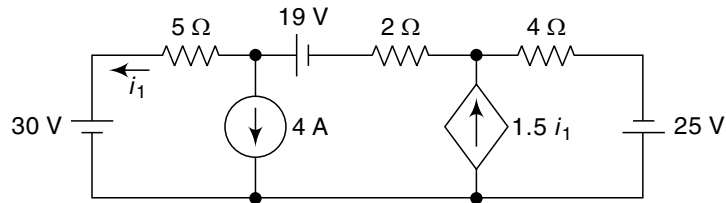
$$\begin{bmatrix} 15 & -5 & -5 \\ -5 & 14 & -5 \\ -5 & -9 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Solving for I_2 ,

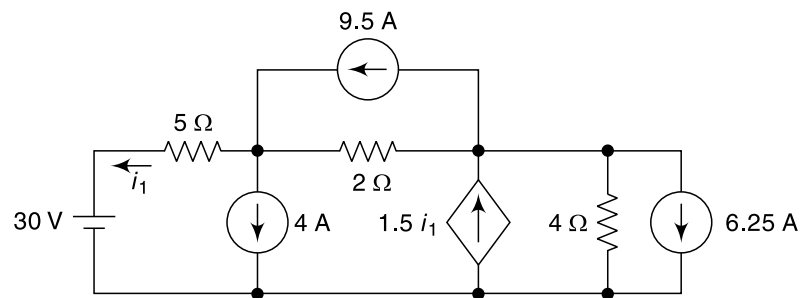
$$I_2 = \frac{\begin{vmatrix} 15 & -1 & -5 \\ -5 & 1 & -5 \\ -5 & 0 & 10 \end{vmatrix}}{\begin{vmatrix} 15 & -5 & -5 \\ -5 & 14 & -5 \\ -5 & -9 & 10 \end{vmatrix}} = \frac{1}{19} \text{ A}$$

$$\therefore V_x = 4 \times I_2 = 4 \times \frac{1}{19} = \frac{4}{19} \text{ V} \quad \text{Ans.}$$

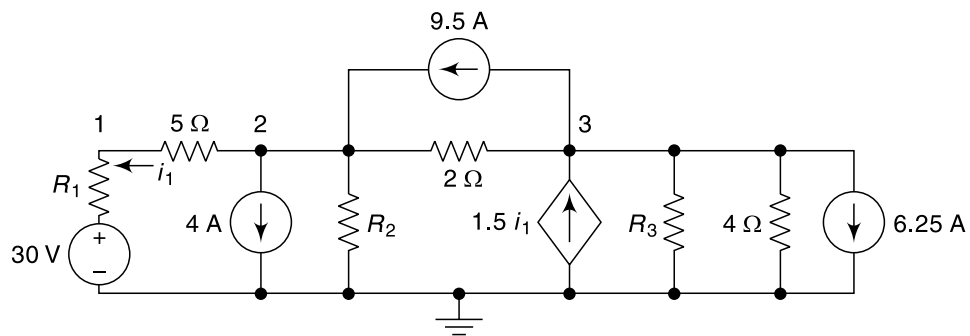
4.9 Determine the current i_1 in the circuit using nodal analysis method and graph theory concepts.



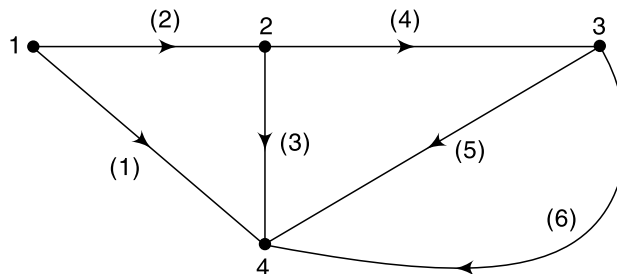
Solution By source transformation technique, we convert the 19 V and 25 V voltage sources into current sources.



Since the 30 V voltage source, the 4 A current source, and controlled current source are not accompanied by the passive elements, we consider three resistors R_1 , R_2 and R_3 and finally let, $R_1 \rightarrow 0$, $R_2 \rightarrow \infty$, and $R_3 \rightarrow \infty$.



The graph of the network is shown.



The complete incidence matrix is,

$$A_a = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 1 \\ 1 & 0 & 0 & 0 & -1 & -1 \end{bmatrix}$$

Reduced Incidence matrix is,

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{bmatrix}$$

Branch admittance matrix is,

$$Y_b = \begin{bmatrix} G_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & G_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \end{bmatrix} \quad \text{where, } G_1 = \frac{1}{R_1}, G_2 = \frac{1}{R_2}, G_3 = \frac{1}{R_3}$$

$$\therefore AY_b = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} G_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & G_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \end{bmatrix}$$

$$= \begin{bmatrix} G_1 & -\frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{5} & G_2 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -1 & -\frac{1}{2} & G_3 & \frac{1}{4} \end{bmatrix}$$

$$\therefore AY_b A^T = \begin{bmatrix} G_1 & -\frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{5} & G_2 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -1 & -\frac{1}{2} & G_3 & \frac{1}{4} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \left(G_1 + \frac{1}{5}\right) & -\frac{1}{5} & 0 \\ -\frac{1}{5} & \left(G_1 + \frac{1}{5} + \frac{1}{2}\right) & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \left(G_3 + \frac{1}{2} + \frac{1}{4}\right) \end{bmatrix}$$

$$AY_b V_s - AI_s = -AI_s = - \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -30G_1 \\ 0 \\ 4 \\ -9.5 \\ -1.5i_1 \\ 6.25 \end{bmatrix} \quad \{\because \text{We made } V_s = 0\}$$

Now,

$$= - \begin{bmatrix} -30G_1 \\ -5.5 \\ (15.75 - 1.5 i_1) \end{bmatrix}$$

Thus, node equations are,

$$\begin{bmatrix} \left(G_1 + \frac{1}{5}\right) & -\frac{1}{5} & 0 \\ -\frac{1}{5} & \left(G_2 + \frac{7}{10}\right) & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \left(G_3 + \frac{3}{4}\right) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 30G_1 \\ 5.5 \\ 1.5 i_1 - 15.75 \end{bmatrix}$$

With $R_1 \rightarrow 0$, $G_1 \rightarrow \infty$, $R_2 \rightarrow \infty$, $G_2 \rightarrow 0$, $R_3 \rightarrow \infty$, $G_3 \rightarrow 0$ the equations become:

$$\begin{aligned} \left(G_1 + \frac{1}{5}\right)V_1 - \frac{1}{5}V_2 &= 30G_1 \\ -\frac{1}{5}V_1 + \left(G_2 + \frac{7}{10}\right)V_2 - \frac{1}{2}V_3 &= 5.5 \\ -\frac{1}{2}V_2 + \frac{3}{4}V_3 &= (1.5i_1 - 15.75) \end{aligned}$$

or

$$V_1 = 30 \quad (i)$$

$$-\frac{1}{5}V_1 + \frac{7}{10}V_2 - \frac{1}{2}V_3 = 5.5 \Rightarrow 7V_2 - 5V_3 = 115 \quad (ii)$$

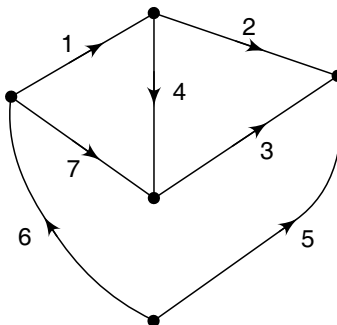
$$-\frac{1}{2}V_2 + \frac{3}{4}V_3 = \left[1.5\left(\frac{V_2 - V_1}{5}\right) - 15.75\right] \Rightarrow 16V_2 - 15V_3 = 495 \quad (iii)$$

Solving equations (i), (ii), and (iii), we get,

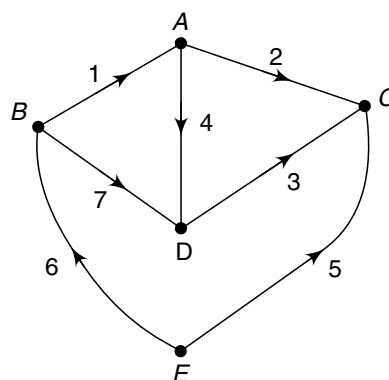
$$V_2 = -30 \text{ V}, \quad V_3 = 65 \text{ V}$$

Hence, the current, $i_1 = \left(\frac{V_2 - V_1}{5} \right) = \frac{-30 - 30}{5} = -12 \text{ A}$ *Ans.*

4.10 Write the complete incidence matrix for the graph shown in the figure.



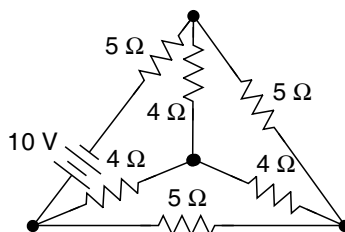
Solution We first label the nodes as shown in the figure.



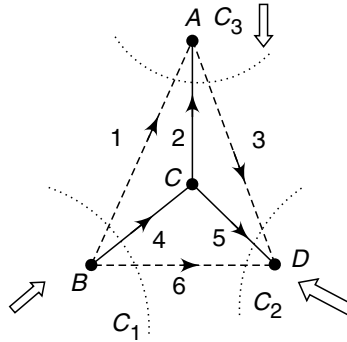
The complete incidence matrix is given as,

	1	2	3	4	5	6	7
A	-1	1	0	1	0	0	0
B	1	0	0	0	0	-1	1
C	0	-1	-1	0	-1	0	0
D	0	0	1	-1	0	0	-1
E	0	0	0	0	1	1	0

4.11 Write down the incidence matrix and cutset matrices for the network shown.



Solution The graph and a suitable tree for the network are shown in the figure.



The complete incidence matrix is given as,

		1	2	3	4	5	6
$A_a =$	A	-1	-1	1	0	0	0
	B	1	0	0	1	0	1
	C	0	1	0	-1	1	0
	D	0	0	-1	0	-1	-1

The fundamental cutsets are identified as,

f-cutset-1: [1, 4, 6]

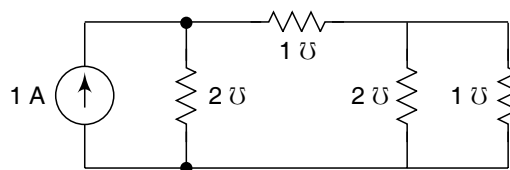
f-cutset-2: [3, 5, 6]

f-cutset-3: [1, 2, 3]

The fundamental cutset matrix is given as,

		1	2	3	4	5	6
$Q =$	C_1	-1	0	0	1	0	1
	C_2	0	0	-1	0	-1	-1
	C_3	-1	-1	1	0	0	0

4.12 For the network shown in the figure, give fundamental cutset matrix and hence find KCL equations.

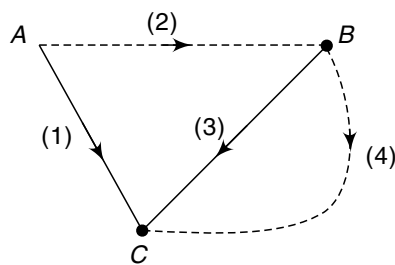


Solution The graph and one tree are shown for the network.

The fundamental cutsets are identified as

f-cutset-1: [1, 2]

f-cutset-2: [2, 3, 4]



The fundamental cutset matrix is given as,

		1	2	3	4
$Q_a =$	C_1	1	1	0	0
	C_2	0	-1	1	1

The KCL equations in terms of the cutset matrix is given as,

$$[Q][Y_b][Q^T][V_t] = -[Q][I_s]$$

Here,

$$[Q][Y_b][Q^T] = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & -1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & -1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

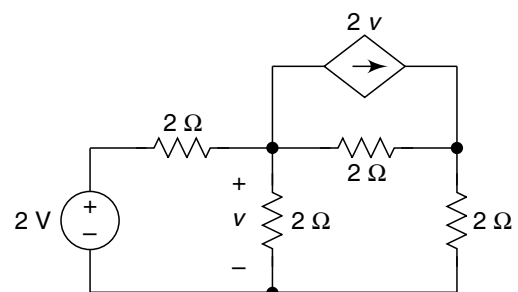
$$= \begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix}$$

$$-[Q][I_s] = -\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

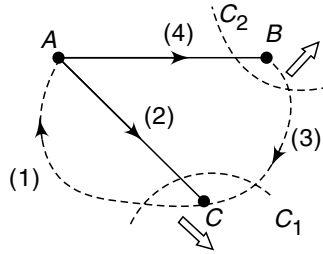
Thus, the KCL equations are

$$\begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} V_{t1} \\ V_{t3} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{Ans.}$$

- 4.13 For the network shown in the figure, draw the oriented graph, select a suitable tree and obtain the fundamental cutset matrix. Determine the node equations and find v .
Solution The oriented graph of the network is shown in the figure. Since we have to find v , we take branch (2) in the twig and a possible tree is selected.



The fundamental cutsets are identified as



f-cutset-1: [1, 2, 3];

f-cutset-2: [3, 4];

The fundamental cutset matrix is given as,

		1	2	3	4
$Q_a =$	C_1	-1	1	1	0
	C_2	0	0	-1	1

The node equations are given as,

$$[Q][Y_b][Q^T][V_t] = [Q] \times \{[Y_b][V_s] - [I_s]\}$$

Here,

$$[Q][Y_b][Q^T] = \begin{bmatrix} -1 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 1 \end{bmatrix}$$

$$[Q] \times \{[Y_b][V_s] - [I_s]\} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 1 \end{bmatrix} \left\{ \begin{bmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 2v \end{bmatrix} \right\} = \begin{bmatrix} 1 \\ -2v \end{bmatrix}$$

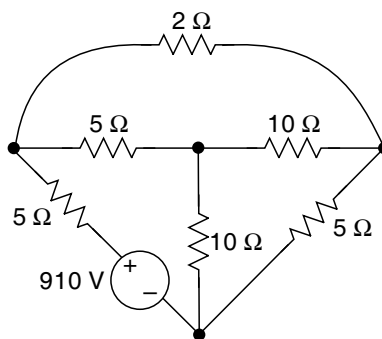
Thus, the KCL equations are

$$\begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 1 \end{bmatrix} \begin{bmatrix} V_{t2} \\ V_{t4} \end{bmatrix} = \begin{bmatrix} 1 \\ -2v \end{bmatrix}$$

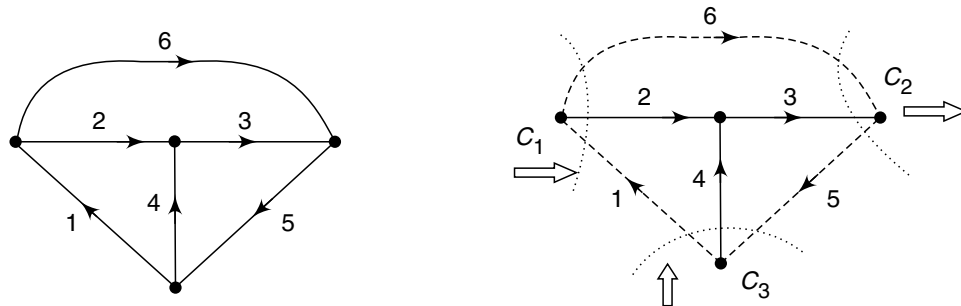
Here, $V_{t2} = v$. Putting this in the KCL equations and solving we get,

$$v = \frac{4}{9}V \quad \text{Ans.}$$

- 4.14 For the resistive network, write a cutset schedule and equilibrium equations on voltage basis. Hence obtain values of branch voltages and branch currents.



Solution The graph of the network is shown in the figure. A suitable tree is shown.



The fundamental cutsets are identified as,

f-cutset-1: [1, 2, 6];

f-cutset-2: [3, 5, 6];

f-cutset-3: [1, 4, 5]

The fundamental cutset matrix is given as,

	1	2	3	4	5	6
C_1	-1	1	0	0	0	1
C_2	0	0	1	0	-1	1
C_3	1	0	0	1	-1	0

The node equations are given as,

$$[Q][Y_b][Q^T][V_t] = [Q] \times \{[Y_b][V_S] - [I_S]\} = [Q][Y_b][V_S] \quad \{\text{since } I_S = 0 \text{ here}\}$$

Here,

$$[Q][Y_b][Q^T] = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 \\ 1 & 0 & 0 & 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1/5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/10 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/10 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0.9 & 0.5 & -0.2 \\ 0.5 & 0.8 & 0.2 \\ 0.5 & 0.2 & 0.3 \end{bmatrix}$$

$$[Q][Y_b][V_s] = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 \\ 1 & 0 & 0 & 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1/5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/10 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/10 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} -910 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 182 \\ 0 \\ 0 \end{bmatrix}$$

Thus, the KCL equations are

$$= \begin{bmatrix} 0.9 & 0.5 & -0.2 \\ 0.5 & 0.8 & 0.2 \\ 0.5 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} V_{t2} \\ V_{t3} \\ V_{t4} \end{bmatrix} = \begin{bmatrix} 182 \\ 0 \\ 0 \end{bmatrix}$$

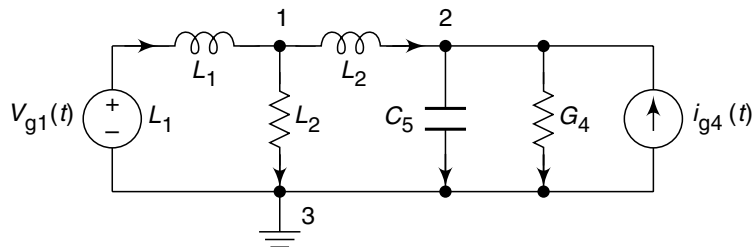
Solving by Cramer's rule, we get the tree-branch voltages as,

$$V_{t2} = 143 \text{ V}; \quad V_{t3} = -14.3 \text{ V}; \quad V_{t4} = -300 \text{ V} \quad \text{Ans.}$$

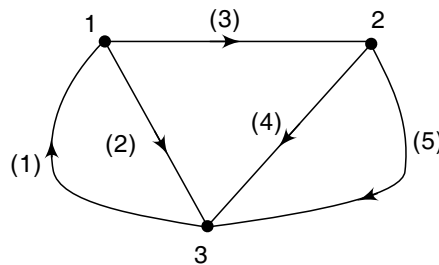
- 4.15 Using topological method, obtain node equations and node voltages in s domain for the network shown in the figure, when

$$L_1 = L_2 = 1 \text{ H}, \quad C_5 = 1 \text{ F}, \quad G_3 = G_4 = 1 \Omega, \quad V_{g1}(t) = 2u(t) \quad \text{and} \quad i_{g4}(t) = 2\delta(t),$$

where, $u(t)$ is unit step function and $\delta(t)$ is the unit impulse function.



Solution The graph of the network is shown in the figure.



The incidence matrix is given as,

	1	2	3	4	5
$A_a =$	1	-1	1	0	0
	2	0	0	1	1
	3	1	-1	-1	-1

Reduced incidence matrix is,

$$A = \begin{bmatrix} -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 \end{bmatrix}$$

Branch admittance matrix is,

$$Y_b = \begin{bmatrix} 1/s & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1/s & 0 & 0 \\ 0 & 0 & 0 & s & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\therefore AY_b = \begin{bmatrix} -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/s & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1/s & 0 & 0 \\ 0 & 0 & 0 & s & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1/s & 1 & 1/s & 0 & 0 \\ 0 & 0 & -1/s & s & 1 \end{bmatrix}$$

$$\therefore AY_b A^T = \begin{bmatrix} -1/s & 1 & 1/s & 0 & 0 \\ 0 & 0 & -1/s & s & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 0 \\ 1 & -1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} (2/s+1) & -1/s \\ -1/s & (s+1+1/s) \end{bmatrix}$$

Now,

$$AY_b V_s - AI_s = \begin{bmatrix} -1/s & 1 & 1/s & 0 & 0 \\ 0 & 0 & -1/s & s & 1 \end{bmatrix} \begin{bmatrix} 2/s^2 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 2/s^2 \\ 2 \end{bmatrix}$$

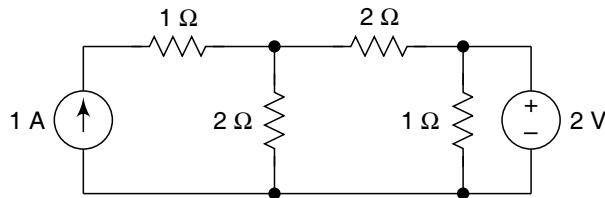
Thus, node equations are,

$$\begin{bmatrix} (2/s+1) & -1/s \\ -1/s & (s+1+1/s) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2/s^2 \\ 2 \end{bmatrix}$$

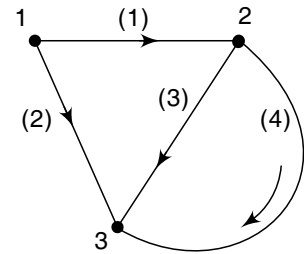
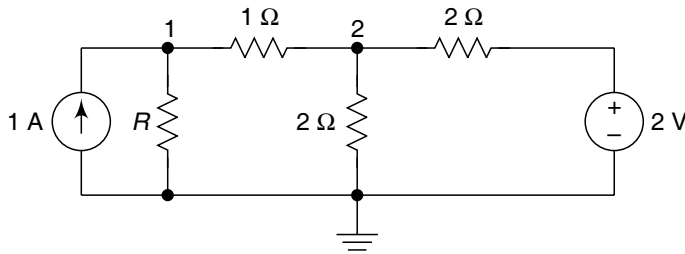
Solving by Cramer's rule, we get the voltages as,

$$V_1 = \frac{2(2s^2 + s + 1)}{s(s+1)(s^2 + 2s + 1)} \quad \text{and} \quad V_2 = \frac{2(s^3 + s^2 + 1)}{s(s+1)(s^2 + 2s + 1)}$$

- 4.16 Determine the currents in all branches of the network shown in the figure using node analysis method. Use graph theory method.



Solution Here, the 1Ω resistance in parallel with the 2 V voltage source can be ignored. Also, there is no passive element in parallel with 1 A current source. We assume a resistance R in parallel with 1 A current source and finally let $R \rightarrow \infty$. Therefore, the graph of the network is shown in the figure.



The complete incidence matrix is,

$$A_a = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 1 \\ 0 & -1 & -1 & -1 \end{bmatrix}$$

Reduced incidence matrix is,

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 1 \end{bmatrix}$$

Branch admittance matrix is,

$$Y_b = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/R & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/2 \end{bmatrix}$$

$$\therefore AY_b = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/R & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 1/R & 0 & 0 \\ -1 & 0 & 1/2 & 1/2 \end{bmatrix}$$

$$\therefore AY_b A^T = \begin{bmatrix} 1 & 1/R & 0 & 0 \\ -1 & 0 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} (1+1/R) & -1 \\ -1 & 2 \end{bmatrix}$$

Now,

$$AY_b V_s - AI_s = \begin{bmatrix} 1 & 1/R & 0 & 0 \\ -1 & 0 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Thus, node equations are,

$$\begin{bmatrix} (1+1/R) & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

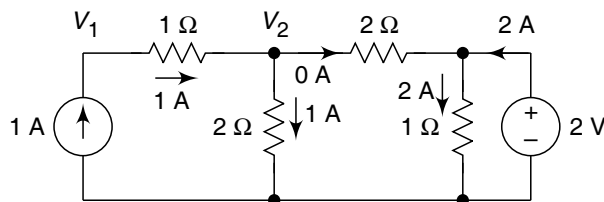
With $R \rightarrow \infty$, the equations become

$$\begin{aligned} V_1 - V_2 &= 1 \\ -V_1 + 2V_2 &= 1 \end{aligned}$$

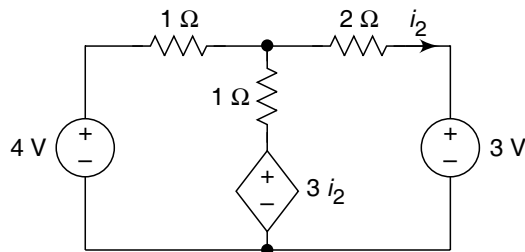
Solving equations, we get,

$$V_1 = 3\text{V}, \quad V_2 = 2\text{V}$$

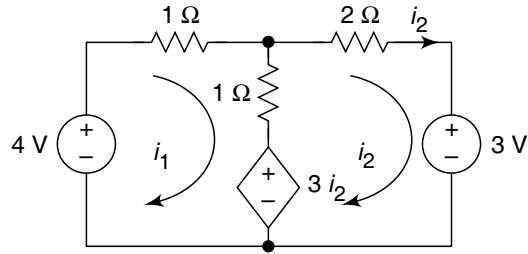
Hence, the currents in different branches are shown in the figure.



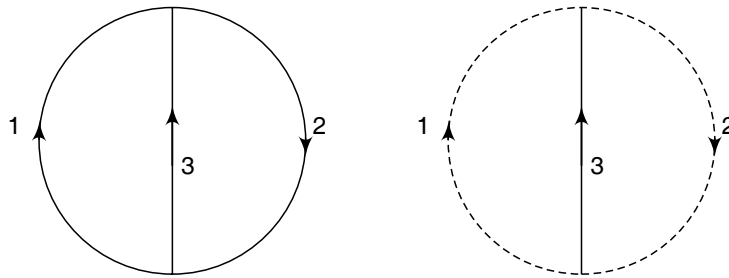
- 4.17 Consider the network shown in the figure using loop method of analysis, determine currents in all the branches, indicating their directions. Use graph theory method.



Solution



The graph of the network is shown below. Also, the tree is selected as shown.



For the selected tree, the tie-set matrix is given as,

$$B_a = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

The branch impedance matrix is,

$$Z_b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore$

$$B_a Z_b = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \end{bmatrix}$$

$\therefore$

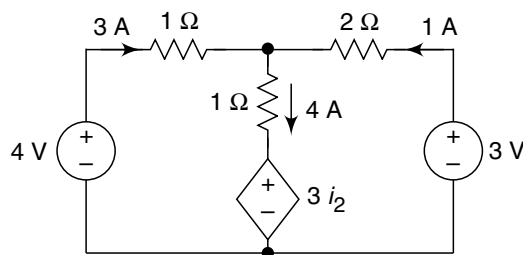
$$B_a Z_b B_a^T = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$$

Now,

$$-B_a V_s = - \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -4 \\ 3 \\ -3i_2 \end{bmatrix} = \begin{bmatrix} 4 - 3i_2 \\ -3 + 3i_2 \end{bmatrix}$$

So, the loop equations become,

$$\begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 4 - 3i_2 \\ -3 + 3i_2 \end{bmatrix}$$



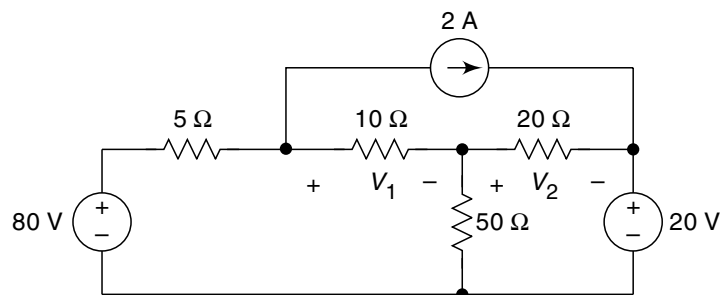
These equations reduce to,

$$\begin{aligned} 2i_1 - i_2 &= 4 - 3i_2 &\Rightarrow i_1 + i_2 &= 2 \\ -i_1 + 3i_2 &= -3 + 3i_2 &\Rightarrow i_1 &= 3A \end{aligned}$$

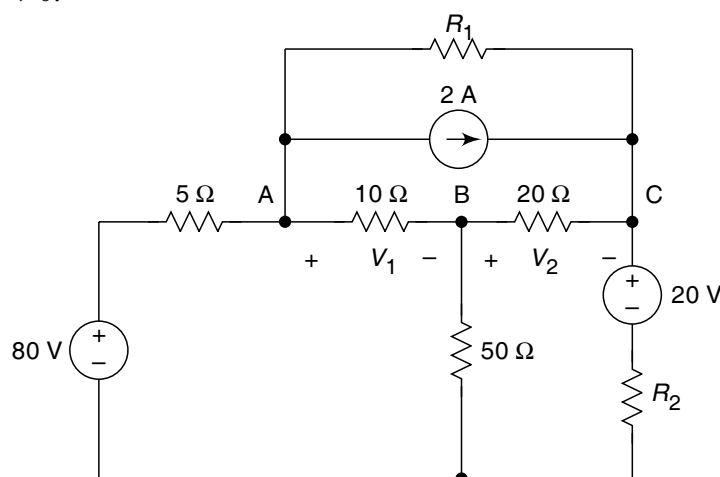
$$\therefore i_2 = -1A$$

Thus, the branch currents are shown with their directions.

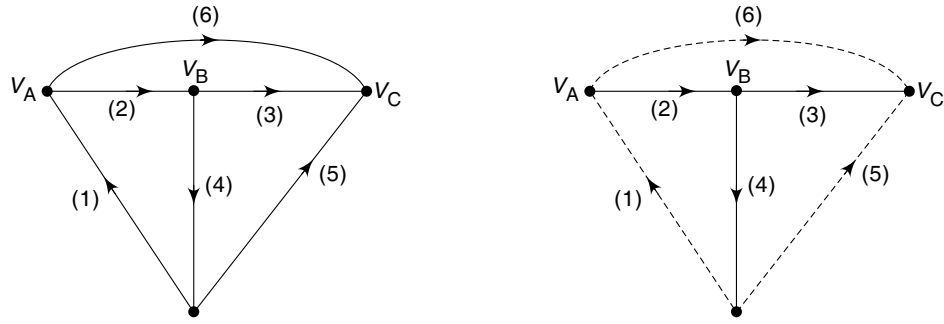
- 4.18 For the circuit shown in the figure construct a tree in which 10Ω and 20Ω are in tree branches. Using node analysis, solve for V_1 and V_2 .



Solution Here, we have on e current source without parallel resistance and one voltage source without series resistance. Therefore, we connect a parallel resistance R_1 in parallel with the 2A current source and a series resistance R_2 in series with the 20V voltage source. Finally, we will let $R_1 \rightarrow \infty$ and $R_2 \rightarrow 0$.



Now, we construct the graph of the network as shown below. A tree, in which 10Ω and 20Ω are in tree branches, is selected.



The complete incidence matrix is,

$$A_a = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & -1 & -1 \\ 1 & 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

Reduced incidence matrix is,

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix}$$

Branch admittance matrix is,

$$Y_b = \begin{bmatrix} 0.2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.05 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.02 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{R_2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{R_1} \end{bmatrix}$$

$\therefore$

$$AY_b = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} 0.2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.05 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.02 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{R_2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{R_1} \end{bmatrix}$$

$$= \begin{bmatrix} -0.2 & 0.1 & 0 & 0 & 0 & \frac{1}{R_1} \\ 0 & -0.1 & 0.05 & 0.02 & 0 & 0 \\ 0 & 0 & -0.05 & 0 & -\frac{1}{R_2} & -\frac{1}{R_1} \end{bmatrix}$$

$$\therefore AY_b A^T = \begin{bmatrix} -0.2 & 0.1 & 0 & 0 & 0 & \frac{1}{R_1} \\ 0 & -0.1 & 0.05 & 0.02 & 0 & 0 \\ 0 & 0 & -0.05 & 0 & -\frac{1}{R_2} & -\frac{1}{R_1} \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} \left(0.3 + \frac{1}{R_1}\right) & -0.1 & -\frac{1}{R_1} \\ -0.1 & 0.17 & -0.05 \\ -\frac{1}{R_1} & -0.05 & \left(0.05 + \frac{1}{R_2} + \frac{1}{R_1}\right) \end{bmatrix}$$

Now,

$$AY_b V_s - AI_s = \begin{bmatrix} -0.2 & 0.1 & 0 & 0 & 0 & \frac{1}{R_1} \\ 0 & -0.1 & 0.05 & 0.02 & 0 & 0 \\ 0 & 0 & -0.05 & 0 & -\frac{1}{R_2} & -\frac{1}{R_1} \end{bmatrix} \begin{bmatrix} -80 \\ 0 \\ 0 \\ 0 \\ -20 \\ 0 \end{bmatrix} - \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 14 \\ 0 \\ \left(\frac{20}{R_2} + 2\right) \end{bmatrix}$$

Thus, node equations are,

$$\begin{bmatrix} \left(0.3 + \frac{1}{R_1}\right) & -0.1 & -\frac{1}{R_1} \\ -0.1 & 0.17 & -0.05 \\ -\frac{1}{R_1} & -0.05 & \left(0.05 + \frac{1}{R_2} + \frac{1}{R_1}\right) \end{bmatrix} \begin{bmatrix} V_A \\ V_B \\ V_C \end{bmatrix} = \begin{bmatrix} 14 \\ 0 \\ \left(\frac{20}{R_2} + 2\right) \end{bmatrix}$$

With $R_1 \rightarrow \infty$, the equations become

$$\begin{aligned} 0.3V_A - 0.1V_B &= 14 \\ -0.1V_A + 0.17V_B - 0.05V_C &= 0 \\ -0.05V_B + \left(0.05 + \frac{1}{R_2}\right)V_C &= \left(\frac{20}{R_2} + 2\right) \end{aligned}$$

Solving equations, we get,

$$V_A = \frac{\begin{vmatrix} 14 & -0.1 & 0 \\ 0 & 0.17 & -0.05 \\ \left(\frac{20}{R_2} + 2\right) & -0.05 & \left(0.05 + \frac{1}{R_2}\right) \end{vmatrix}}{\begin{vmatrix} 0.3 & -0.1 & 0 \\ -0.1 & 0.17 & -0.05 \\ 0 & -0.05 & \left(0.05 + \frac{1}{R_2}\right) \end{vmatrix}} = \frac{14[0.17(0.05R_2 + 1) - 0.0025R_2] + 0.005(20 + 2R_2)}{0.3[0.17(0.05R_2 + 1) - 0.0025R_2] - 0.01(0.05R_2 + 1)}$$

With $R_2 = 0$,

$$V_A = \frac{2.48}{0.041} = 60.49 \text{ V}$$

Similarly, with $R_2 = 0$, we get,

$$V_B = 41.47 \text{ V}$$

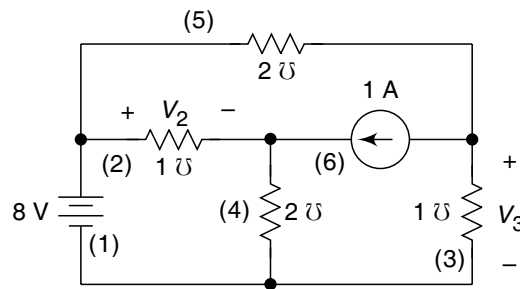
$$V_C = 20 \text{ V}$$

$\therefore$

$$V_1 = (V_A - V_B) = (60.49 - 41.47) = 19.02 \text{ V} \quad \text{Ans.}$$

$$\text{and } V_2 = (V_B - V_C) = (41.420) = 21.47 \text{ V} \quad \text{Ans.}$$

- 4.19 In the following circuit, determine the voltages V_2 and V_3 using cutset analysis. Select the circuit elements (1), (2) and (3) in the tree.



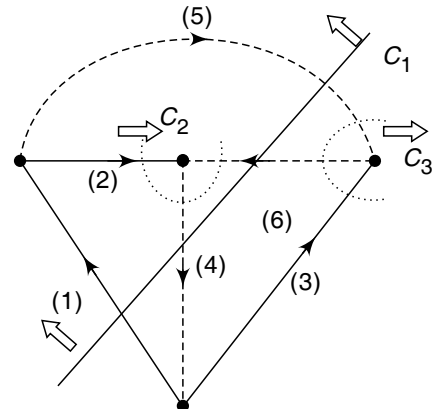
Solution The graph and tree are shown in the figure. Hence, there is no series impedance with voltage source and parallel admittance with current source. We consider two resistances R_1 and R_2 in series with the voltage source and in parallel with the current source, respectively. Finally, we will let $R_1 \rightarrow 0$, $R_2 \rightarrow \infty$.

Three fundamental cutsets are:

f-cutset-1: [1, 4, 5, 6];

f-cutset-2: [2, 4, 6];

f-cutset-3: [3, 5, 6]



The fundamental cutset matrix is given as,

	1	2	3	4	5	6
$Q =$						
C_1	1	0	0	-1	-1	1
C_2	0	1	0	1	0	-1
C_3	0	0	1	0	1	1

The node equations are given as,

$$[Q][Y_b][Q^T][V_t] = [Q] \times \{[Y_b][V_s] - [I_s]\}$$

Here,

$$\begin{aligned}
 [Q][Y_b][Q^T] &= \begin{bmatrix} 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/R_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/R_2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} (4 + 1/R_1 + 1/R_2) & -(2 + 1/R_2) & -(2 + 1/R_2) \\ -(2 + 1/R_2) & -(3 + 1/R_2) & 1/R_2 \\ -(2 + 1/R_2) & 1/R_2 & (3 + 1/R_2) \end{bmatrix} \\
 [Q][Y_b][V_s] &= \begin{bmatrix} 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/R_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/R_2 \end{bmatrix} \begin{bmatrix} -8 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -8/R_1 \\ 0 \\ 0 \end{bmatrix} \\
 -[Q][I_s] &= -\begin{bmatrix} 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}
 \end{aligned}$$

Thus, the KCL equations are

$$\begin{bmatrix} (4 + 1/R_1 + 1/R_2) & -(2 + 1/R_2) & -(2 + 1/R_2) \\ -(2 + 1/R_2) & -(3 + 1/R_2) & 1/R_2 \\ -(2 + 1/R_2) & 1/R_2 & (3 + 1/R_2) \end{bmatrix} \begin{bmatrix} V_{t1} \\ V_{t2} \\ V_{t3} \end{bmatrix} = \begin{bmatrix} -(8/R_1 + 1) \\ 1 \\ 1 \end{bmatrix}$$

When $R_1 \rightarrow 0$, $R_2 \rightarrow \infty$, the equations reduce to the form as given below.

$$V_{t1} = -8 \text{ volts}$$

$$-2V_{t1} - 3V_{t2} = 1$$

$$-2V_{t1} + 3V_{t3} = 1$$

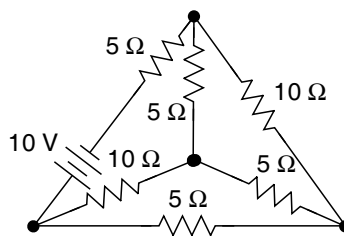
Solving the last two equations, $V_{t2} = 5$ volts; $V_{t3} = -5$ volts

Therefore,

$$V_2 = V_{t2} = 5 \quad \text{Ans.}$$

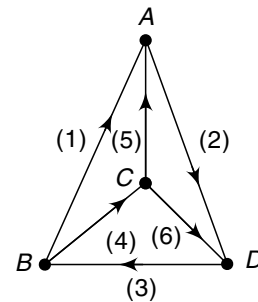
$$V_3 = -V_{t3} = 5 \quad \text{Ans.}$$

- 4.20 For the network shown in the figure, write the tie-set matrix and determine the loop currents and branch currents.



Solution The graph and a suitable tree for the network are shown in the figure. The tie-set matrix is given as,

$$B_a = \begin{bmatrix} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{bmatrix}$$



The branch impedance matrix is given as,

$$Z_b = \begin{bmatrix} 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 10 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 \end{bmatrix}$$

$$\therefore B_a Z_b = \begin{bmatrix} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 10 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 & 10 & -5 & 0 \\ 0 & 10 & 0 & 0 & 5 & -5 \\ 0 & 0 & 5 & -10 & 0 & 5 \end{bmatrix}$$

$$\therefore B_a Z_b B_a^T = \begin{bmatrix} 5 & 0 & 0 & 10 & -5 & 0 \\ 0 & 10 & 0 & 0 & 5 & -5 \\ 0 & 0 & 5 & -10 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 20 & -5 & -10 \\ -5 & 20 & -5 \\ -10 & -5 & 15 \end{bmatrix}$$

$$-B_a V_s = - \begin{bmatrix} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} -10 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

Thus, the loop equations are given as,

$$\begin{bmatrix} 20 & -5 & -10 \\ -5 & 20 & -5 \\ -10 & -5 & 15 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

Solving by Cramer's rule, we get the loop currents as,

$$I_1 = \frac{\begin{vmatrix} 10 & -5 & -10 \\ 0 & 20 & -5 \\ 0 & -5 & 15 \end{vmatrix}}{\begin{vmatrix} 20 & -5 & -10 \\ -5 & 20 & -5 \\ -10 & -5 & 15 \end{vmatrix}} = \frac{2750}{2625} = 1.047 \text{ A}$$

$$I_2 = \frac{\begin{vmatrix} 20 & 10 & -10 \\ -5 & 0 & -5 \\ -10 & 0 & 15 \end{vmatrix}}{\begin{vmatrix} 20 & -5 & -10 \\ -5 & 20 & -5 \\ -10 & -5 & 15 \end{vmatrix}} = \frac{1250}{2625} = 0.476 \text{ A}$$

$$I_3 = \frac{\begin{vmatrix} 20 & -5 & 10 \\ -5 & 20 & 0 \\ -10 & -5 & 0 \end{vmatrix}}{\begin{vmatrix} 20 & -5 & -10 \\ -5 & 20 & -5 \\ -10 & -5 & 15 \end{vmatrix}} = \frac{2250}{2625} = 0.857 \text{ A}$$

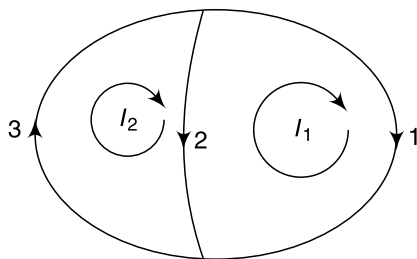
Also, the branch currents are given as, $I_b = B_a^T I_L$

$$\therefore \begin{bmatrix} I_{b1} \\ I_{b2} \\ I_{b3} \\ I_{b4} \\ I_{b5} \\ I_{b6} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1.047 \\ 0.476 \\ 0.857 \end{bmatrix}$$

$$\therefore \left. \begin{aligned} I_{b4} &= (1.047 - 0.857) = 0.19 \text{ A} \\ I_{b5} &= (-1.047 + 0.476) = -0.571 \text{ A} \\ I_{b6} &= (-0.476 + 0.857) = 0.381 \text{ A} \end{aligned} \right\} \text{ Ans.}$$

MULTIPLE-CHOICE QUESTIONS

- 4.1 The number of links for a graph having n nodes and b branches are
 (a) $b - n + 1$ (b) $n - b + 1$ (c) $b + n - 1$ (d) $b + n$
- 4.2 The tree branches of a graph are called
 (a) chords (b) links (c) twigs (d) co-tree
- 4.3 The tie-set matrix gives the relation between
 (a) branch currents and link currents. (b) branch voltages and link currents.
 (c) branch currents and link voltages. (d) none of these.
- 4.4 The graph of a network has six branches with three tree branches. The minimum number of equations required for the solution of the network is
 (a) 2 (b) 3 (c) 4 (d) 5
- 4.5 For a connected planar graph of v vertices and e edges, the number of meshes is
 (a) $(e - v + 1)$ (b) $(e + v + 1)$ (c) $(e + v - 1)$
- 4.6 The number of chords of a tree of a connected graph G of v vertices and e edges is
 (a) $(v - 1)$ (b) $(e - v + 1)$ (c) $(e - v - 1)$
- 4.7 The table meant for the oriented graph represents



Link or Loop Current	← Branch →		
	1	2	3
i_1	+1	-1	0
i_2	0	+1	+1

- (a) tie-set matrix (b) cut-set matrix (c) incidence matrix (d) none of the above.

4.8 The reduced incidence matrix of a circuit is given by

$$A_i = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{bmatrix} 1 & -1 & -1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

The set of branches forming a tree are

- (a) 1, 2 and 3 (b) 2, 3 and 5 (c) 1, 2 and 4 (d) 1, 2 and 6.

4.9 Relative to a given fixed tree of a network

1. link currents form an independent set.
2. branch currents form an independent set.
3. link voltages form an independent set.
4. branch voltages form an independent set.

Of these statements

- (a) 1, 2, 3 and 4 are correct. (b) 1, 2 and 3 are correct.
 (c) 2, 3 and 4 are correct (d) 1, 3 and 4 are correct

4.10 For a given network the incidence matrix is given as

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & -1 & 1 \\ -1 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

The series branches in the graph are

- (a) 3 and 4 (b) 6 and 7 (c) 2 and 3 (d) none of the above.

4.11 For a given network the incidence matrix is given as

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & -1 & 1 \\ -1 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

The parallel branches in the graph are

- (a) 1 and 2 (b) 2 and 3 (c) 6 and 7 (d) none of the above.

4.12 For a given network the incidence matrix is given as

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

The series branches in the graph are

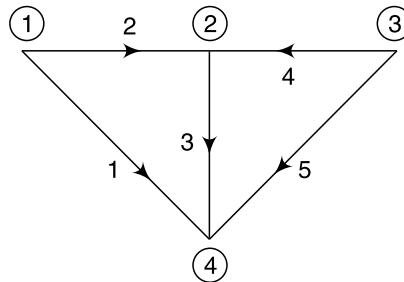
- (a) 3 and 4 (b) 3 and 5 (c) 3 and 6 (d) none of the above.

4.13 For a given network the incidence matrix is given as

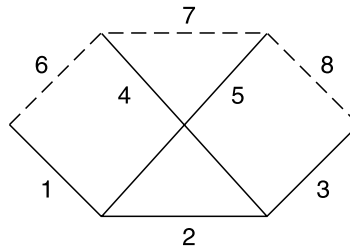
$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

The parallel branches in the graph are

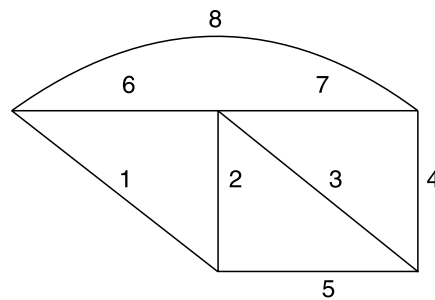
- (a) 3 and 5 (b) 4 and 5 (c) 3 and 6 (d) none of the above.
- 4.14 Which one of the following represents the total number of trees in the graph given in the figure?



- (a) 4 (b) 5 (c) 6 (d) 8
- 4.15 In the graph and the tree shown in the given figure, the fundamental cut-set for the branch 2 is



- (a) 2, 1, 5 (b) 2, 6, 7, 8 (c) 2, 1, 3, 4, 5 (d) 2, 3, 4
- 4.16 In the graph shown in the figure, one possible tree is formed by the branches 4, 5, 6, 7. Then one possible fundamental cut set is



- (a) 1, 2, 3, 8 (b) 1, 2, 5, 6 (c) 1, 5, 6, 8 (d) 1, 2, 3, 7, 8
- 4.17 Which one of the following statements is correct?
A tree in a network is a connected graph containing
- (a) all the nodes only
(b) all the branches only

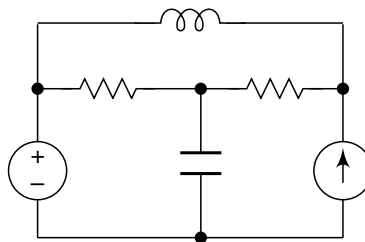
- (c) all the branches and nodes
 (d) all the nodes but no close path

4.18

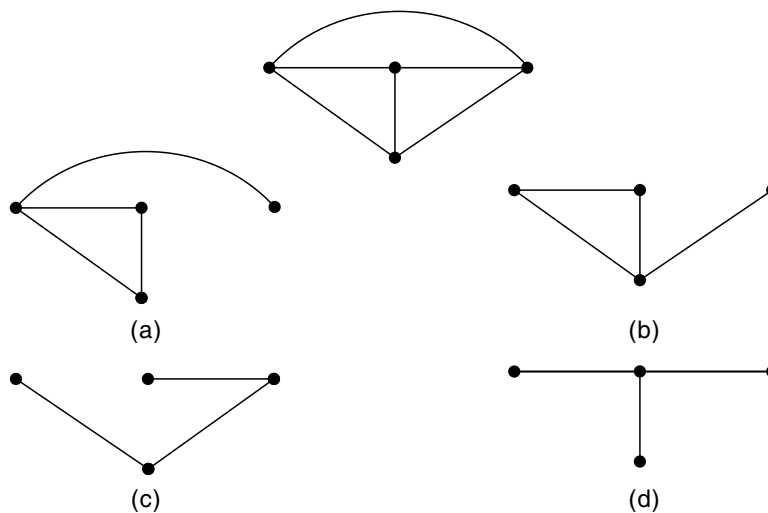
$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & -1 & -1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

For the reduced incidence matrix given, which is the set of branches forming a tree?

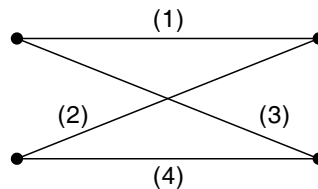
- (a) 1, 2, 3 (b) 2, 4, 6 (c) 2, 3, 5 (d) 1, 4, 6
- 4.19 The number of chords in the graph of the given circuit will be



- (a) 3 (b) 4 (c) 5 (d) 6
- 4.20 Consider the network graph shown in the figure. Which one of the following is NOT a 'tree' of this graph?



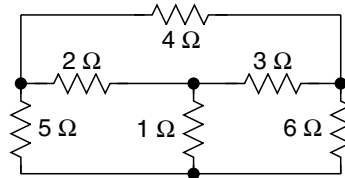
- 4.21 In the following graph, the number of trees (P) and the number of cut-sets (Q) are



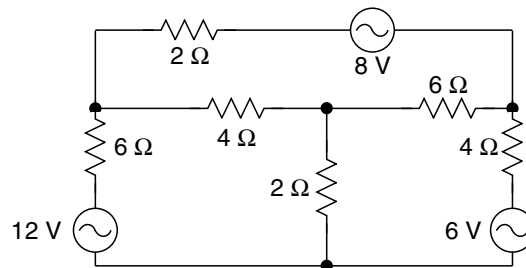
- (a) $P = 2, Q = 2$ (b) $P = 2, Q = 6$ (c) $P = 4, Q = 6$ (d) $P = 4, Q = 10$

EXERCISES

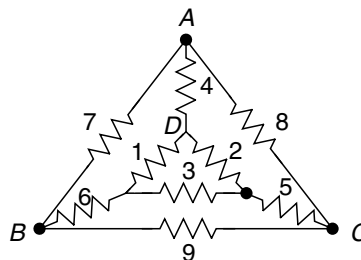
- 4.1 For the network shown in the figure, draw the graph and a possible tree. Show the links and write the tie-set matrix. Write the equations of the branch currents in terms of loop currents.



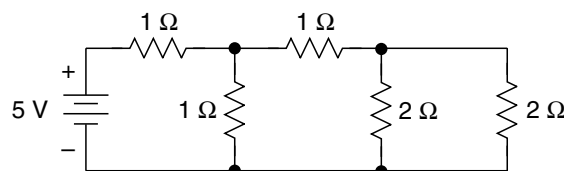
- 4.2 Find out the currents through and voltage across all branches of the network shown in the figure with the help of its tie-set schedule.



- 4.3 Find a tree from the graph of the network shown in the figure. Make the tie-set matrix and write the equations containing branch currents and loop currents. All the values are in ohms.

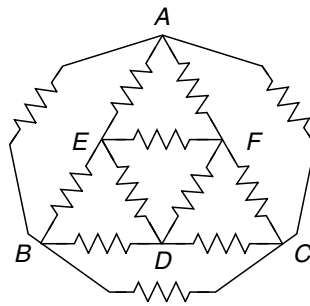


- 4.4 Draw the graph of the circuit shown in the figure and select a suitable tree to write tie-set matrix. Then find the three loop currents.

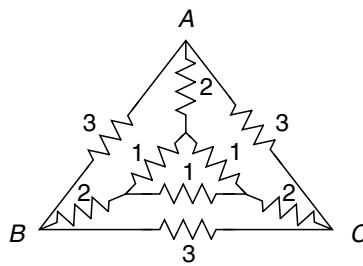


$$[i_1 = 3A, i_2 = 1A, i_3 = 0.5A]$$

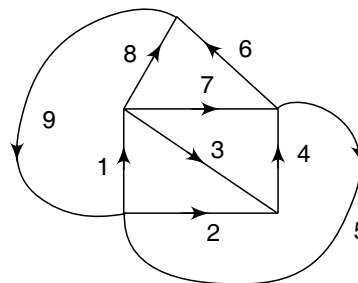
- 4.5 For the given network of the figure, draw the graph and a tree. Write the cutsets and the cutset matrix of the tree. Write the equations of link branch voltages in terms of tree branch voltages.



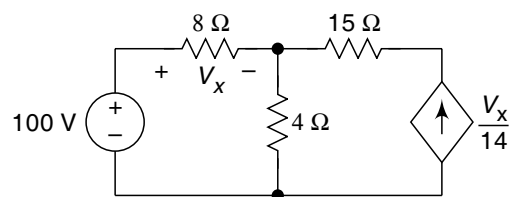
- 4.6 For the given network of the figure, draw the graph and a tree. Write the cutsets and the cutset matrix of the tree. Write the equations of link branch voltages in terms of tree branch voltages. All the values are in ohms.



- 4.7 The linear oriented graph is given in the figure. Considering a tree, mark all the fundamental cutsets and form the cutset matrix.



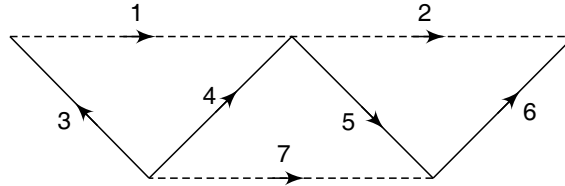
- 4.8 For the network shown, determine



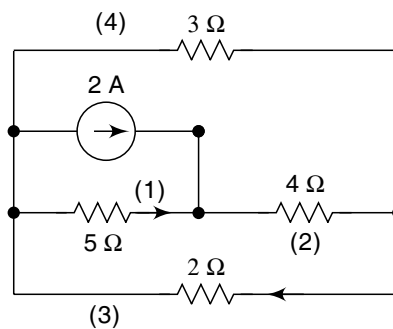
- tie-set matrix,
- loop impedance matrix, and
- loop currents.

[7A, 4A]

- 4.9 Select the (i) fundamental loops, and (ii) fundamental cutsets corresponding to a tree of the network graph which is shown by solid lines in the figure. Hence write the KCL and KVL equations for the network in matrix form.



- 4.10 Draw the graph of the network. Select a tree with tree branches of elements (1) and (2) and write the equilibrium equation taking tree branch voltages as variables.



- 4.11 The incidence matrix of a network graph is given below. Draw the oriented graph.

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 \end{bmatrix}$$

SHORT-ANSWER TYPE QUESTIONS

- 4.1 Give the topological description of networks.
- 4.2 (a) Define the following terms:
- (i) Graph of a network
 - (ii) Oriented graph
 - (iii) Rank of graph
 - (iv) Planar and non-planar graph
 - (v) Subgraph
 - (vi) Path
- (b) State the advantages offered by the graph theory as applied to electric circuit problems.
- 4.3 What is meant by graph? How does a graph help in circuit analysis?
- 4.4 (a) Define a tree of a graph of a network. Mention some basic properties of a 'Tree'. How can you calculate the number of possible trees of a given graph?
- (b) Define the followings
- (i) Twigs.
 - (ii) Cotree
 - (iii) Links or chords

- 4.5 Show that the number of links for a graph having n nodes and b branches is $b - n + 1$.
- 4.6 Show that for a network graph with P separate parts, n nodes and b branches, the number of chords C is given as, $C = b - n + P$.
- 4.7 Explain with illustrative examples the meaning of the following terms:
- Incidence matrix
 - Tie-set matrix
 - Cut-set matrix
- 4.8 (a) Explain what is meant by incidence matrix of a graph and indicate how the values of the incidence matrix elements are obtained.
- List the properties of an incidence matrix.
 - How can you determine the number of possible trees of a graph with this matrix?
- 4.9 Show that the determinant of the complete incidence matrix of a closed loop is zero.
- 4.10 (a) Explain the term 'tie-set' and 'tie-set matrix' of a network with an illustrative example.
- Show that the matrix equation,

$$I_b = B^T I_L$$

where, B is the tie-set matrix and I_b and I_L represent branch current and loop current matrix respectively.

- Write the tie-set schedule and formulate the equilibrium equation on loop current basis.
- 4.11 (a) Define cut-set in a network graph. How can you find out a fundamental cut-set? Mention some properties of a cut-set.
- Define cut-set matrix with an illustrative example and show that the matrix equation $QI_b = 0$, where Q is the cut-set matrix and I_b represents the branch current matrix of the graph.
 - Briefly discuss the relation between branch voltage matrix and node voltage matrix in terms of cut-sets.
- 4.12 Prove that in a linear graph, every cut-set has an even number of branches in common with every loop.
- 4.13 (a) Write notes on network equilibrium equation.
- Establish that the independent loop equations of a network can be formulated from the tie-set matrix of its graph, with an illustrative example.
 - Establish the formulation of node equations of a network from the cut-set matrix.
- 4.14 Using the topological properties of a network graph, describe the step-by-step procedure of analysing a network by node voltage method.
- 4.15 Using the topological properties of a network graph, describe the step-by-step procedure of analysing a network by loop current method.

ANSWERS TO MULTIPLE-CHOICE QUESTIONS

4.1 (a)	4.2 (c)	4.3 (a)	4.4 (b)	4.5 (a)	4.6 (b)	4.7 (a)
4.8 (a)	4.9 (b)	4.10 (d)	4.11 (c)	4.12 (c)	4.13 (b)	4.14 (d)
4.15 (b)	4.16 (d)	4.17 (d)	4.18 (a)	4.19 (a)	4.20 (b)	4.21 (c)